Supplemental Table 1. Search strings for respective databases (Search date: 2021/7/29).

Research step	Query	Study number
PubMed (MEDLINE)		
1	antibiotic [MeSH] OR antimicrobial [MeSH] OR antibiotic [Title/Abstract] OR antimicrobial [Title/Abstract] OR antibiotic [Title/Abstract] OR antimicrobial [Title/Abstract]	942,233
2	appropriate [MeSH] OR inappropriate [MeSH] OR adequate [MeSH] OR inadequate [MeSH] OR appropriate [Title/Abstract] OR inappropriate [Title/Abstract] OR adequate [Title/Abstract] OR inadequate [Title/Abstract]	964,057
3	empirical [MeSH] OR initial [MeSH] OR empirical [Title/Abstract] OR initial [Title/Abstract]	959,975
4	bacteremia [MeSH] OR bacteraemia [MeSH] OR bloodstream [MeSH] OR bacteremia [Title/Abstract] OR bacteraemia [Title/Abstract] OR bloodstream [Title/Abstract]	320,688
5	mortality [MeSH] OR fatality [MeSH] OR death [MeSH] OR dead [MeSH] OR alive [MeSH] OR survival [MeSH] OR mortality [Title/Abstract] OR fatality [Title/Abstract] OR death [Title/Abstract] OR dead [Title/Abstract] OR alive [Title/Abstract] OR survival [Title/Abstract]	2,531,989
6	Children[Title/Abstract] OR neonate[Title/Abstract] OR adolescent[Title/Abstract] OR infant[Title/Abstract] OR pediatric[Title/Abstract]	1.586,707
7	#1 AND #2 AND #3 AND #4 AND #5 NOT #6	579
Cochrane		
1	MeSH descriptor: [antibiotic OR antimicrobial] explode all text	1,644
2	MeSH descriptor: [empirical OR initial OR appropriate OR inappropriate OR adequate OR inadequate] explode all text	8,206
3	MeSH descriptor: [bacteremia OR bacteraemia OR bloodstream] explode all text	407
4	MeSH descriptor: [mortality OR fatality OR death OR dead OR alive OR survival] explode all text	6,295
5	MeSH descriptor: [children OR neonate OR adolescent OR infant OR pediatric] explode title, abstract and keyword	2,813
6	#1 AND #2 AND #3 AND #4 NOT #5	204
Embase		
1	“antibiotic”/all fields OR “antimicrobial” /all fields	935,364
2	“appropriate” /all fields OR “inappropriate” /all fields OR “adequate” /all fields OR “inadequate” /all fields	134,870
3	“empirical”/all fields OR “initial”/all fields	1,294,873
4	“bacteremia" /all fields OR “bacteraemia”/all fields OR “bloodstream” /all fields	14,662
5	“mortality" /all fields OR “fatality” /all fields OR “death” /all fields OR “dead” /all fields OR “alive”/all fields OR”	1,214,168

	survival”/all fields	
6	“children”/ti,ab OR “neonate”/ti,ab OR “adolescent”/ti,ab OR “infant”/ti,ab OR “pediatric”/ti,ab	2,026,479
7.	#1 AND #2 AND #3 AND #4 AND #5 NOT #6	963

Supplemental Table 2. All included studies.

Study and country	Multi-center/Prospective/cohort	Study population /Patient No.	Acquisition place/Bacteraemia source	Specific population	Microorganisms; antibiotic-R isolates (%)	Primary outcome, (mortality rate)	Cut-off timeline for EAT (after initial culture)	Patient No. (%) of appropriate EAT	OR (95% C.I.) of inappropriate EAT for mortality	AOR (95% C.I.) of inappropriate EAT for mortality	Quality (NOQ assessment)
Bodey et al, 1986; US [203]	No/no/yes	Hospital/621	-/overall	Hemato-oncology	*E. coli*	7-day crude, 28.8	0 h	559 (90.0)	ND	NA	5
Meyers et al, 1989; US [202]	No/no/yes	Hospital/100	-/overall	the elderly	All	In-hospital crude, 40.0	0 h	84 (84.0)	ND	NA	5
Phillips et al, 1990; UK [200]	No/no/yes	Hospital/1505	-/overall	Overall	All	In-hospital crude, 18.7	0 h	1346 (89.4)	ND	NA	5
Feldman et al, 1990; South African [201]	No/yes/yes	Hospital/47	-overall	Overall	*K. pneumoniae*	In-hospital crude, 55.3	0 h	41 (87.2)	ND	NA	4
Elhanan et al, 1997; Israel [199]	No/yes/yes	Hospital/134	Community/UTI	Overall	All	In-hospital crude, 12.7	0 h	124 (92.5)	0.63 (0.08-5.24)	ND, NS	6
Salonen et al, 1998; Finland [196]	No/no/yes	Hospital/57	-/overall	Overall	Anaerobes	In-hospital crude, 24.6	BC result	28 (49.1)	ND	NA	5
Leibovici et al, 1998; Israel [197]	No/yes/yes	Hospital/3413	-/overall	Overall	All	In-hospital crude, 25.4	48h	2158 (63.2)	2.1 (1.8-2.4)	1.6 (1.3-1.9)	7
Carratala et al, 1998; Spain [198]	No/yes/yes	Hospital/40	-/pneumonia	Hemato-oncology	All	30-day crude, 62.5	BC result	35 (87.5)	ND	NA	6
Byl et al, 1999; Belgium [195]	No/yes/yes	Hospital/428	-/overall	Overall	All	In-hospital crude, 19.9	24h	269 (62.9)	ND	0.47 (0.25-0.87)	7
Soriano et al, 2000; Spain [193]	No/no/yes	Hospital/908	-/overall	Overall	*S. aureus*; MRSA (24.8)	7-day sepsis, 12.1	BC result	719 (79.2)	ND	2.13 (1.21-3.75)	7
Ibrahim et al, 2000; US [194]	No/yes/yes	ICU/492	-/overall	Overall	All	In-hospital crude, 38.4	BC result	345 (70.1)	2.18 (1.77-2.69)	6.86 (5.09-9.24)	6
Hanon et al, 2002; Denmark [192]	No/no/yes	Hospital/2058	-/overall	Overall	All	180-day crude, 30.6	0 h	1290 (62.7)	ND	0.64 (0.48-0.84)	7
Kang et al, 2003; South Korea [190]	No/no/yes	Hospital/136	-/overall	Overall	*P. aeruginosa*	30-day crude, 39.0	24h	47 (34.6)	2.14 (1.00-4.58)	4.61 (1.18-18.09)	7
Chamot et al, 2003; Switzerland [191]	No/no/yes	Hospital/115	-/overall	Overall	*P. aeruginosa*	30-day crude, 13.9	24h	98 (85.2)	1.1 (0.29-4.5)	1.2 (0.29-5.2)	8
Zaragoza et al, 2003; Spain [186]	No/yes/yes	ICU/166	-/overall	Overall	All	In-hospital crude, 51.8	BC result	127 (76.5)	ND	ND, NS	6
Valles et al, 2003; Spain [187]	Yes/yes/yes	ICU/339	Community/overall	Overall	All	in-hospital crude, 41.6	24h	290 (85.5)	ND	3.23 (1.52-6.83)	7
Endimiani et al, 2003; Italy [3]	No/no/yes	Hospital/521	-/overall	Overall	All	In-hospital crude, 11.3	BC result	314 (60.3)	ND	4.28 (0.68-72.08)	7
Lodise et al, 2003; US [189]	No/no/yes	Hospital/167	Nosocomial/overall	Overall	*S. aureus*; MRSA (61.7)	In-hospital sepsis, 23.4	44.75h	119 (71.3)	ND	3.8 (1.3-11.0)	7
Mackenzie et al, 2003; UK [188]	No/no/yes	Hospital/78	-/overall	Overall	All	In-hospital crude, 20.0	BC result	54* (83.1)	ND	NA	7
Lin et al, 2004; Taiwan [183]	No/no/yes	Hospital/86	Community/overall	Overall	*S. aureus*; MRSA (25.6)	14-day crude, 25.6	0 h	70 (81.4)	ND	NA	7
Anatoliotaki et al, 2004; Greece [185]	No/no/no	Hospital/157	-/overall	Oncology	All	In-hospital sepsis, 19.7	24h	111 (70.7)	ND	0.25 (ND)	7
Bouza et al, 2004; Spain [184]	No/yes/yes	Hospital/297	-/overall	Overall	All	In-hospital crude, 23.6	24h	170 (58.6)	0.5 (0.3-0.9)**	NA	6
Hung et al, 2005; Taiwan [181]	No/yes/yes	Hospital/52	Community/overall	Overall	Anaerobes	30-day crude, 25.0	72h	26* (74.3)	ND	NA	6
Bouza et al, 2005; Spain [182]	Yes/no/yes	Hospital/77	Nosocomial/overall	Overall	*S. pneumoniae*	In-hospital crude, 45.5	24h	65 (84.4)	ND	10.6 (1.2-97.0)	8
Shih et al, 2005; Taiwan [177]	No/no/yes	Hospital/71	-/overall	Overall	*S. marcescens*	14-day crude, 40.8	48h	30 (42.3)	ND	NA	7
Wang et al, 2005; Taiwan	No/no/yes	Hospital/371	-/overall	Hematology	All	In-hospital crude, 25.1	0 h	303 (81.7)	16.80 (8.60-33.15)	NA	6

[176]											
Micek,et al, 2005; US [178]	No/no/yes	Hospital/305	-/overall	Overall	*P. aeruginosa*	In-hospital crude, 21.0	0 h	230 (75.4)	ND	2.04 (1.42-2.92)	8
Kang et al, 2005; South Korea [180]	No/no/yes	Hospital/286	-/overall	Overall	GNB; 3rd GC-R (100)	30-day crude, 33.2	24h	135 (47.2)	1.65 (1.001-2.73)	ND,NS	8
Metan et al, 2005; Turkey [179]	No/no/yes	Hospital/53	-/overall	Overall	*E. coli*; ESBL (100)	30-day crude, 26.4	0 h	12 (22.6)	3.00 (0.86-10.41)	NA	7
Ye et al, 2006; China [167]	No/no/yes	Hospital/126	-/overall	Overall	*Enterobacter* spp.; MDR (15.2)	30-day crude, 39.7	5d	58 (46.0)	ND	ND, NS	7
Schramm et al, 2006; US [168]	No/no/yes	Hospital/559	-/overall	Overall	*S. aureus*; MRSA (100)	In-hospital crude, 23.1	24h	169 (30.2)	ND	1.92 (1.48-2.50)	8
Garey et al, 2006; US [173]	Yes/no/yes	Hospital/230	Nosocomial/overall	Overall	*Candida* spp.	In-hospital crude, 24.5	24h	130* (67.7)	ND	1.50 (1.09-2.09)	8
Falagas et al, 2006; Greece [175]	No/no/yes	Hospital/40	-/overall	Overall	*A. baumannii*	in-hospital crude, 47.5	72h	18 (45.0)	ND	NA	7
Marra et al, 2006; US [169]	No/no/yes	Hospital/91	-/overall	Overall	*S. aureus*; MRSA (46.2)	In-hospital crude, 14.3	24h	64 (70.3)	2.3 (0.70-7.75)	NA	7
Khatib et al, 2006; US [171]	No/yes/yes	Hospital/342	-/overall	Overall	*S. aureus*; MRSA (50.9)	In-hospital crude, 23.7	0 h	282 (82.5)	ND	ND, NS	7
Kim et al, 2006; South Korea [4]	No/no/yes	Hospital/238	-/overall	Overall	*S. aureus*; MRSA (53.4)	12-week sepsis, 33.2	48h	121 (50.8)	1.60 (0.93-2.76)	1.39 (0.62-3.15)	9
Fang et al, 2006; Taiwan [174]	No/no/yes	Hospital/162	Nosocomial/overall	Overall	*S. aureus*; MRSA (100)	30-day crude, 36.4	48h	43 (26.5)	ND	0.83 (0.66-1.04)	7
Lin et al, 2006; Taiwan [170]	No/no/yes	Hospital/108	-/overall	Overall	*E. cloacae*	In-hospital sepsis, 20.4	0 h	67 (62.0)	ND	NA	6
Guilarde et al 2006; Brazil [172]	No/no/yes	Hospital/111	-/overall	Overall	*S. aureus*; MRSA (60.2)	In-hospital sepsis, 35.1	48h	67 (60.4)	2.90 (1.44-5.84)	2.27 (1.02-5.09)	7
Peralta et al, 2007; Spain [162]	No/no/yes	Hospital/663	-/overall	Overall	*E. coli*; ESBL (3.3)	In-hospital crude, 5.4	24h	557 (84.0)	2.6 (1.34-5.03)	2.98 (1.25-7.11)	6
Ortega et al, 2007; Spain [164]	No/yes/yes	Hospital/200	Community/primary	Overall	All	In-hospital crude, 13.0	BC result	162 (81.0)	ND	2.0 (1.22-3.33)	7
Bassetti et al, 2007; Italy [166]	Yes/yes/yes	Hospital/136	-/overall	Overall	*Candida* spp.	21-day crude, 55.1	72h	87* (73.7)	4.89 (1.71-15.87)	4.75 (1.71-13.20)	6
Osih et al, 2007; US [163]	No/no/yes	Hospital/167	-/overall	Overall	*P. aeruginosa*	In-hospital crude, 36.5	24h	99 (59.3)	ND	NA	7
Lodise et al, 2007; US [165]	No/no/yes	Hospital/100	-/overall	Overall	*P. aeruginosa*	30-day crude, 31.0	52h	52 (52.0)	ND	4.1 (1.2-13.9)	6
Labelle AJ et al, 2008; US [160]	No/no/yes	Hospital/245	-/overall	Overall	*Candida* spp.	In-hospital crude, 45.7	24h	37 (15.1)	ND	NA	7
Cheong et al, 2008; South Korea [161]	No/no/yes	Hospital/614	Community/overall	Overall	*P. aeruginosa*, *E. coli*	30-day crude, 15.8	24h	508 (82.7)	ND	3.51 (1.17-10.52)	6
Tumbarello et al, 2008; Italy [155]	No/no/yes	Hospital/129	-/overall	Overall	*E. coli*; ESBL (100)	21-day crude, 29.4	BC result	73 (56.6)	7.63 (2.97-20.58)	6.22 (2.33-16.61)	6
Marcos et al, 2008; Spain [158]	No/no/yes	Hospital/377	-/overall	Overall	*Enterobacter* spp.	30-day crude, 12.5	BC result	232 (61.6)	ND	NA	6
Marschall et al, 2008; US [157]	No/yes/yes	Hospital/250	-/overall	Overall	GNB	In-hospital crude, 14.0	24h	171 (68.4)	ND	NA	7
Wareham et al, 2008; UK [154]	No/no/yes	Hospital/399	-/overall	Overall	*Acinetobacter* spp.; MDR (34.0)	30-day crude,6.3	BC result	88* (54.3)	ND	NA	7
Soriano et al, 2008; Spain [156]	No/yes/yes	Hospital/414	-/overall	Overall	*S. aureus*; MRSA (100)	30-day crude, 28.0	BC result	168 (40.6)	0.8 (0.5-1.3)**	NA	6
Lin et al, 2008; US [159]	No/no/yes	Hospital/1523	-/overall	Overall	All	30-day crude, 8.5	24h	983 (64.5)	ND	NA	9
Su et al, 2009; Taiwan [145]	No/no/yes	Hospital/123	-/overall	Overall	*Fusobacterium* spp.	In-hospital crude, 40.7	BC result	66 (53.6)	ND	NA	7
Erbay et al, 2009; Turkey [150]	No/no/yes	Hospital/103	Nosocomial/overall	Overall	*A. baumannii*	30-day crude, 54.4	48h	43 (41.7)	ND	2.4 (1.3-4.2)	8

Ammerlaan et al, 2009;Europe [153]	Yes/no/yes	Hospital/334	-/overall	Overall	*S. aureus*; MRSA (23.1)	30-day crude, 24.0	BC result	240 (71.9)	0.69 (0.36-1.32)	NA	7
Metan et al, 2009; Turkey [147]	No/yes/yes	Hospital/100	Nosocomial/overall	Overall	*A. baumannii*; MDR (48.0)	14-day crude, 63.0	24h	24 (24.0)	ND	ND, NS	8
Evans et al, 2009; US [149]	No/no/yes	Hospital/235	-/overall	Spinal cord injury	All	30-day crude, 6.8	48h	147 (62.6)	ND	1.74 (0.61-4.94)	8
Klevay et al, 2009; US [148]	Yes/yes/yes	Hospital/322	-/overall	Overall	*Candida* spp.	28-day crude, 29.8	0 h	244 (75.8)	ND	NA	6
Daikos et al, 2009; Greece [151]	Yes/yes/yes	Hospital/162	Nosocomial/overall	Overall	*K. pneumoniae*; MBL (41.4)	14-day crude, 19.1	48h	133 (82.1)	2.27 (0.91-5.64)	1.63 (0.68-3.86)	9
Ortega et al, 2009; Spain [146]	No/yes/yes	Hospital/4758	-/overall	Overall	*E. coli;* ESBL (4.4)	30-day crude, 9.2	BC result	4409 (92.7)	ND	4.83 (3.48-6.71)	7
Trecarichi et al, 2009; Italy [144]	No/no/yes	Hospital/62	-/overall	Hematology	*E. coli;* ESBL (41.9)	30-day crude, 21.0	48h	51 (82.3)	5.40 (2.25-12.96)	14.96 (1.95-114.51)	8
Chang et al, 2009; Taiwan [152]	No/no/yes	Hospital/88	-/overall	Overall	*Enterobacter aeruginosa*	In-hospital sepsis, 10.2	BC result	55 (62.5)	ND	ND, NS	6
Abhilash et al, 2010; India [143]	No/yes/yes	Hospital/131	-/overall	Overall	*E. coli, K. pneumoniae*; ESBL (73.3)	14-day crude, 23.6	0 h	74 (56.5)	0.76 (0.34-1.73)**	NA	6
Khan et al, 2010; Qatar [141]	Yes/yes/no	Hospital/452	-/overall	Overall	All	In-hospital crude, 22.5	24h	387 (85.6)	ND	8.35 (3.26-29.05)	7
Vitkauskiene et al, 2010; Lithuania [130]	No/no/yes	Hospital/80	-/overall	Overall	*P. aeruginosa*	30-day crude, 28.8	BC result	49 (61.3)	1.05 (0.42-2.61)**	NA	6
Paul et al, 2010; Israel [137]	No/no/yes	Hospital/510	-/overall	Overall	*S. aureus*; MRSA (100)	30-day crude, 43.9	48h	168 (32.9)	ND	2.15 (1.34 - 3.46)	8
Lin et al, 2010; Taiwan [140]	No/no/yes	Hospital/40	-/vascular catheter	Overall	*Chryseobacterium* spp.	14-day crude, 52.5	72h	17 (42.5)	4.19 (1.10-15.90)	ND, S	7
Tumbarello et al, 2010; Italy [131]	No/no/yes	Hospital/134	-/overall	Overall	*E. coli*; ESBL (27.6)	21-day crude, 12.7	0 h	107 (79.9)	3.09 (1.92-4.98)	NA	7
Schweizer et al, 2010; US [134]	No/no/yes	Hospital/814	-/overall	Overall	*S. aureus*; MRSA (60.0)	30-day crude, 22.9	24h	537 (66.0)	1.52 (0.99-2.34)	1.50 (0.96-2.35)	9
Son et al, 2010; South Korea [133]	Yes/yes/yes	Hospital/1144	Healthcare-associated/overall	Overall	All	30-day crude, 13.9	24h	930 (81.3)	ND	6.04 (2.1616.87)	8
Martinez et al, 2010; Spain [139]	No/yes/yes	Hospital/4863	-/overall	Overall	GNB	30-day crude, 9.6	BC result	4409 (90.7)	2.2 (1.7-2.9)	1.8 (1.3-2.5)	7
Tam et al, 2010; US [132]	No/no/yes	Hospital/109	-/overall	Overall	*P. aeruginosa*; MDR (22.9)	30-day crude, 18.3	24h	93 (85.3)	0.295 (0.093-0.943)**	ND, NS	8
Corona et al, 2010; UK [142]	Yes/yes/yes	ICU/1702	-/overall	Overall	All	In-ICU crude, 38.7	0 h	689 (40.5)	0.79 (0.58-1.07)**	0.89 (0.61-1.30)**	7
Rodriguez-Bano et al, 2010; Spain [136]	Yes/yes/yes	Hospital/96	Nosocomial/overall	Overall	*E. coli;* ESBL (100)	30-day crude, 30.2	24h	53 (55.2)	1.0 (0.5-1.8)	ND, NS	7
Micek et al, 2010; US [138]	No/no/yes	Hospital/760	-/overall	Overall	GNB	In-hospital crude, 47.0	24h	238 (31.3)	ND	2.30 (1.89-2.80)	8
Rodriguez-Bano J et al, 2010; Spain [135]	Yes/no/yes	Hospital/282	Community/overall	Overall	*E. coli*; ESBL (33.7)	14-day crude, 11.0	24h	205 (72.7)	3.3 (1.6-6.2)	3.0 (1.3-12.2)	7
Tuon et al, 2011; Brazil [110]	No/no/yes	Hospital/104	-/overall	Overall	*K. pneumoniae*; ESBL (58.7)	30-day crude, 46.2	48h	55 (52.9)	ND	NA	6
Rosa et al, 2011; Italy [127]	No/no/yes	Hospital/128	-/overall	Overall	*E. coli, K. pneumoniae, P. mirabilis*; ESBL (100)	21-day crude,17.2	72h	74 (81.3	ND	NA	7
Rebelo et al, 2011; Portugal [117]	No/no/yes	Hospital/135	-/overall	the elderly	All	In-hospital crude, 22.2	BC result	46 (34.1)	0.375 (0.139-1.010)**	ND, NS	8
Feodoroff et al, 2011;	Yes/no/yes	Population/76	-/overall	Overall	*Campylobacter*	30-day crude, 2.6	0 h	30 (39.5)	ND	NA	6

Finland [125]7					spp.						
Ku et al, 2011; South Korea [122]	No/no/yes	Hospital/27	-/overall	Overall	*Burkholderia cepacia*	28-day crude, 40.7	24h	19 (70.4)	ND	23.921 (1.313-435.967)	7
Park et al, 2011; South Korea [118]	No/no/yes	Hospital/150	Community/overall	Overall	*E. coli;* ESBL (23.3)	30-day crude, 11.3	24h	108 (72.0)	2.0 (0.7-5.5)	NA	7
Schechner et al, 2011; Israel [115]	Yes/yes/yes	Hospital/76	-/overall	Overall	*P. aeruginosa*	In-hospital crude, 35.5	24h	44 (57.9)	1.7 (0.9-3.2)	ND, NS	8
Wang et al, 2011; Taiwan [109]	No/no/yes	Hospital/113	-/overall	Overall	*E. coli, K. pneumoniae*; ESBL (100)	14-day crude, 23.9	BC result	50 (44.2)	ND	NA	7
Ortega et al, 2011; Spain [119]	No/yes/yes	Hospital/910	-/overall	Overall	*K. pneumoniae*; ESBL (12.3)	30-day crude, 10.5	BC result	824 (90.5)	ND	ND, NS	7
Shime et al, 2011; Japan [114]	No/no/yes	Hospital/310	-/overall	Immunocompetent	All	In-hospital crude, 7.1	24h	270 (87.1)	ND	NA	6
Johnson et al, 2011; US [123]	No/no/yes	Hospital/754	-/overall	Overall	GNB	In-hospital crude, 41.1	24h	519 (68.8)	ND	2.03 (1.66-2.49)	8
Asgeirsson et al, 2011; lceland [129]	No/no/yes	Hospital/325	-/overall	Overall	*S. aureus*	30-day crude, 13.8	BC result	262 (80.6)	ND	NA	7
Reisfeld et al, 2011; Israel [116]	No/no/yes	Hospital/378	-/overall	Dementia	All	30-day crude, 38.6	48h	227 (60.1)	1.72 (1.13-2.62)	1.41 (0.86-2.29)	9
Song et al, 2011; South Korea [113]	No/no/yes	Hospital/28	-/overall	Overall	*A. baumannii*; carbapenem-R (100)	30-day crude, 53.6	48h	13 (46.4)	ND	2.1 (0.2-23.3)	7
Tumbarello et al, 2011; Italy [111]	Yes/no/yes	Hospital/106	-/overall	Overall	*P. aeruginosa*; MDR (37.7)	21-day crude, 34.0	BC result	69 (65.1)	3.89 (1.52-9.99)	2.73 (1.08-6.85)	7
Gudiol et al, 2011; Spain [124]	No/yes/no	Hospital/372	-/overall	Hemato-oncology	GNB; MDR (13.7)	30-day crude, 22.0	BC result	290* (83.1)	ND	1.57 (0.50-4.90)	7
Kang et al, 2011; South Korea [128]	Yes/no/no	Hospital/365	-/IAI	Overall	GNB; 3rd-GC-R (19.5)	30-day crude, 11.5	24h	301 (82.5)	ND	NA	7
Lewis et al, 2011; UK [120]	No/no/yes	Hospital/195	-/overall	Overall	*S. aureus*; MRSA (100)	30-day crude, 26.7	BC result	63 (32.3)	ND	NA	6
Enoch et al, 2011; UK [126]	No/yes/yes	Hospital/203	-/overall	Overall	GNB; ESBL (3.0)	30-day crude, 22.2	0 h	163 (80.3)	2.63 (1.09-6.34)	NA	7
Lee et al, 2011; South Korea [121]	No/no/yes	Hospital/164	Community/UTI	Overall	All	In-hospital crude, 1.2	BC result	135 (82.3)	0 (0-0)	NA	7
Takesue et al, 2011; Japan [112]	No/no/yes	Hospital/128	-/overall	Overall	*S. aureus*; MRSA (100)	28-day crude, 34.4	BC result	22 (17.2)	ND	0.22 (0.06-0.91)**	7
Zarkotou et al, 2011; Greece [108]	No/yes/yes	Hospital/53	-/overall	Overall	*K. pneumoniae*; KPC (100)	In-hospital crude, 34.0	48h	14 (26.4)	1.11 (0.31-3.99)	NA	7
Horino et al, 2012; Japan [106]	No/no/yes	Hospital/134	-/overall	Overall	*P. aeruginosa*	30-day crude, 20.9	24h	71 (53.0)	ND	NA	6
Park et al, 2012; South Korea [98]	No/no/yes	Hospital/100	-/pneumonia	Overall	*P. aeruginosa*	28-day crude, 51.0	48h	65 (65.0)	ND	2.73 (1.11-6.71)	7
Peralta et al, 2012; Spain [97]	Yes/no/yes	Hospital/387	-/overall	Overall	*E. coli, K. pneumoniae;* ESBL (100)	In-hospital crude, 20.9	24h	198 (51.2)	0.69 (0.47-1.02)**	0.39 (0.31-0.97)**	7
Jung et al, 2012; South Korea [105]	Yes/no/yes	Hospital/553	Community/overall	Overall	*K. pneumoniae*; ESBL (3.3)	30-day crude, 17.5	BC result	513 (92.8)	2.46 (1.22-4.96)	2.43 (1.07-5.52)	8
Tumbarello et al, 2012; Italy [92]	Yes/no/yes	Hospital/125	-/overall	Overall	*K. pneumoniae*; KPC (100)	30-day crude, 41.6	0h	50 (40.0)	2.00 (1.129-3.34)	4.17 (1.61-10.76)	8
Morata et al, 2012; Spain [100]	No/yes/yes	Hospital/709	-/overall	Overall	*P. aeruginosa*; MDR (17.5)	30-day crude, 19.9	24h	463 (65.3)	ND	2.18 (1.215-3.899)	7

Labelle A et al, 2012; US [104]	No/no/yes	Hospital/906	-/overall	Overall	All	In-hospital crude, 60.2	24h	436 (48.1)	ND	NA	7
Ortega et al, 2012; Spain [99]	No/yes/yes	Hospital/1373	-/BTI	Overall	All	30-day crude, 9.2	BC result	1242 (90.5)	ND	1.4 (1.1-1.7)	7
Lee CC et al, 2012; Taiwan [103]	No/no/yes	ED/454	-/overall	Overall	All	28-day crude, 9.3	0h	330 (72.7)	1.36 (1.0-1.8)	2.26 (1.01-5.13)	8
Sancho et al, 2012; Spain [94]	No/yes/yes	ICU/371	Nosocomial/overall	Overall	All	In-hospital crude, 59.3	BC result	216 (58.2)	ND	1.93 (0.96 - 3.86)	7
Wu et al, 2012; Taiwan [91]	No/yes/yes	Hospital/62	-/overall	Overall	*E. coli*; ESBL (100)	30-day crude, 21.0	48h	38 (61.3)	1.61 (0.50-5.24)	NA	6
Retamar et al, 2012; Spain [96]	Yes/yes/yes	Hospital/801	-/overall	Overall	All	14-day crude, 18.5	BC result	602 (75.2)	1.47 (1.10-2.02)	2.12 (1.34-3.34)	7
Bassetti et al, 2012; Italy [107]	Yes/yes/yes	Hospital/165	-/overall	Overall	*S. aureus*; MRSA (53.9)	30-day crude, 21.2	BC result	109 (66.1)	4.07 (1.74-9.60)	2.79 (1.19-6.52)	7
Lye et al, 2012; Singapore [101]	Yes/no/no	Hospital/675	Nosocomial/overall	Overall	GNB; MDR (44.6)	30-day crude, 22.2	24h	381 (56.4)	0.67 (0.46-0.96)**	0.87 (0.55-1.38)**	8
Lee YT et al, 2012; Taiwan [102]	No/no/yes	Hospital/252	-/overall	Overall	*A. baumannii*	14-day crude, 29.8	48h	91 (36.1)	0.24 (0.12-0.47)**	0.22 (0.10-0.50)**	7
Rong et al, 2012; China [95]	No/no/yes	Hospital/118	Nosocomial/overall	Overall	*S. aureus*; MRSA (63.5)	In-hospital crude, 28.0	48h	72 (61.0)	ND	0.25 (0.09-0.69)**	7
Tumbarello M et al, 2012, Italy [93]	No/no/yes	Hospital/99	-/overall	Overall	*Proteus* spp.; MDR (36.4)	21-day crude, 30.3	BC result	78 (78.8)	14.62 (4.10-57.81)	9.85 (2.67-36.25)	7
Kuo et al, 2013; Taiwan [82]	No/no/yes	Hospital/256	Nosocomial/overall	Overall	*Acinetobacter* spp.	14-day crude, 9.4	48h	135 (50.8)	0.89 (0.39-2.02)**	NA	6
Horcajada et al, 2013; Spain [85]	Yes/yes/yes	Hospital/667	-/UTI	Overall	All	30-day crude, 10.2	24h	540 (81.0)	1.2 (0.6-2.2)	NA	6
Yang et al, 2013; Taiwan [75]	No/no/yes	Hospital/222	-/overall	Overall	All	In-hospital crude, 46.8	24h	91 (41.0)	2.250 (1.297-3.904)	3.715 (1.736-7.948)	7
Gudiol et al, 2013; Spain [86]	No/yes/yes	Hospital/92	-/overall	Hemato-oncology	*Enterococcus faecium*; VRE (0)	30-day crude, 28.3	BC result	59 (64.1)	ND	NA	6
Retamar et al, 2013; Spain [5]	Yes/yes/yes	Hospital/341	-/overall	Overall	All	30-day crude, 20.8	24h	284 (83.3)	1.6 (1.0-2.5)	2.00 (0.95-4.34)	7
Bang et al, 2013; South Korea [90]	No/no/yes	Hospital/61	-/overall	Overall	*P. aeruginosa*	30-day crude, 19.7	24h	18 (29.5)	ND	NA	6
Ortega et al, 2013; Spain [79]	No/yes/yes	Hospital/1007	-/UTI	Overall	All	30-day crude, 9.1	BC result	643* (71.9)	ND	1.86 (1.48-2.44)	7
Ruiz-Giardin et al, 2013; Spain [76]	No/no/yes	Hospital/323	-/overall	Overall	All	30-day crude, 11.8	BC result	226* (76.1)	2 (1.01-4.25)	NA	6
Frakking et al, 2013; Netherland [89]	Yes/no/yes	Hospital/232	-/overall	Overall	*E. coli, K. pneumoniae, Enterobacter* spp.; ESBL (100)	30-day crude, 19.9	24h	85 (36.6)	ND	NA	6
Pena et al, 2013; Spain [77]	Yes/yes/yes	Hospital/593	-/overall	Overall	*P. aeruginosa*	30-day crude, 29.7	24h	332 (56.0)	1.72 (1.02-2.90)	1.70 (0.99-2.92)	7
Kang, Wi et al, 2013; South Korea [83]	No/no/yes	Hospital/92	Community/overall	Overall	*E. coli*; ESBL (100)	30-day crude, 10.9	0 h	28 (30.4)	ND	NA	6
Gasch et al, 2013; Spain [87]	Yes/yes/yes	Hospital/579	-/overall	Overall	*S. aureus*; MRSA (100)	2-day crude, 8.5	48h	378 (65.3)	4.33	3.59 (1.63-7.89)	7
Chen et al, 2013; Taiwan [9]	No/no/yes	ED/937	Community/overall	Overall	All	30-day crude, 17.0	24h	682 (72.8)	ND	1.77 (1.29-2.44)	7
Kang et al, 2013; South Korea [84]	No/no/yes	Hospital/556	-/BTI	Overall	All	30-day crude, 8.8	24h	299 (53.8)	ND	2.25 (1.13-4.48)	7
Molina et al, 2013; Spain [80]	No/yes/yes	Hospital/210	Nosocomial/overall	Overall	All	30-day crude, 9.5	0 h	65 (31.0)	ND	ND, NS	7
Garnacho-Montero et al, 2013; Spain [88]	No/yes/yes	Hospital/188	-/vascular catheter	Overall	Candida spp.	In-hospital crude, 36.7	48h	66 (35.1)	0.66 (0.37-1.20)**	0.40 (0.23-0.83)**	7
Park et al, 2013; South	Yes/no/yes	Hospital/149	Nosocomial/overall	Overall	*P. aeruginosa, A.*	14-day crude, 29.5	72h	73 (49.0)	ND	0.23 (0.07-0.78)**	7

Korea [78]					*baumannii*						
Lee et al, 2013; Taiwan [81]	No/no/yes	Hospital/130	-/overall	Overall	*A. baumannii*; carbapenem-R (20.8)	30-day crude, 40.8	0 h	67 (51.5)	ND	0.31 (0.15-0.72)**	6
Zilberberg et al, 2014; US [63]	No/no/yes	Hospital/1064	-/overall	Overall	GNB; MDR (5.9)	In-hospital crude, 29.2	24h	819 (77.0)	ND	3.872 (2.770-5.413)	8
Shorr et al, 2014; US [64]	No/no/yes	ICU/131	-/overall	Overall	*A. baumannii;* carbapenem-R (58.0)	In-hospital crude, 49.6	24h	38 (29.0)	ND	1.418 (1.099-1.583)	8
Falcone et al, 2014; Italy [71]	No/yes/yes	Hospital/94	-/overall	Overall	*E. coli*, *K. pneumoniae*, *Enterobacter* spp.; ESBL (100)	21-day crude, 37.2	72h	52 (55.3)	ND	NA	6
Rodriguez-Pardo et al, 2014; Spain [65]	No/yes/yes	Hospital/291	Nosocomial/vascular catheter	Overall	All	30-day crude, 14.5	48h	157 (54.0)	0.43 (0.21-0.86)**	0.36 (0.17-0.77)**	7
Hernandez et al, 2014; Spain [69]	No/yes/yes	Hospital/745	Community/primary	Overall	All	30-day crude, 13.4	24h	556 (74.6)	ND	NA	6
Davis et al, 2014; Australia [72]	No/yes/yes	Hospital/41	Community/pneumonia	Overall	*A. baumannii*	28-day crude, 9.8	24h	41 (100)	ND	NA	6
Marin et al, 2014; Spain [66]	No/yes/yes	Hospital/528	-/overall	Oncology	All	30-day crude, 30.9	BC result	114 (21.6)	ND	NA	6
Bartoletti et al, 2014; Italy [74]	No/no/no	Hospital/162	-/overall	Liver cirrhosis	All	30-day crude, 29.0	24h	100 (61.7)	ND	2.92 (1.50-5.41)	8
Kim et al, 2014; South Korea [68]	No/no/yes	Hospital/234	-/overall	Overall	P. aeruginosa; MDR (2.6)	14-day crude, 22.2	24h	196 (83.8)	ND	NA	7
Lee et al, 2014; Taiwan [67]	No/no/yes	ICU/298	Nosocomial/overall	Overall	*A. baumannii;* MDR (31.5)	30-day crude, 33.6	48h	177 (59.4)	9.91 (5.66-17.36)	7.06 (3.59-13.87)	7
Bodro et al, 2014; Spain [73]	No/yes/yes	Hospital/1148	-/overall	Hemato-oncology	All	30-day crude, 22.0	BC result	880 (75.7)	ND	NA	6
Girometti et al, 2014 Italy [70]	No/no/yes	Hospital/217	-/overall	Overall	*K. pneumoniae*; ESBL (23.0)	30-day crude, 26.7	BC result	93 (42.9)	ND	1.9 (1.1-3.4)	7
Al-Dorzi et al, 2015; Saudi Arabia [62]	No/no/yes	ICU/60	Nosocomial/overall	Overall	*A. baumannii*	In-hospital crude, 90.0	BC result	36 (60.0)	ND	NA	6
Lee et al, 2015; Taiwan [57]	No/no/yes	Hospital/189	-/overall	Overall	*S. aureus*; MRSA (100)	30-day crude, 29.1	48h	148 (78.3)	9.91 (5.66-17.36)	6.78 (2.61-17.60)	7
Picot-Gueraud et al, 2015; France [54]	No/no/yes	Hospital/228	-/overall	Overall	Enterobacteriaceae; MDR (100)	30-day crude, 23.0	24h	137 (60.1)	ND	NA	6
Boel et al, 2015; Denmark [6]	No/no/yes	Hospital/2008	-/overall	Overall	All	30-day crude, 19.5	24h	1702 (84.8)	0.65 (0.51-0.83)**	0.79 (0.62-1.01)**	7
Park H et al, 2015; South Korea [56]	No/no/yes	Hospital/102	-/overall	Liver cirrhosis	All	30-day crude, 30.4	0 h	54 (52.9)	2.00 (0.97-4.13)	3.24 (1.50-7.00)	8
Park SY et al, 2015; South Korea [55]	No/no/yes	Hospital/109	-/overall	Overall	Coagulase(-) staphylococci; methicillin-R (86.2)	30-day crude, 23.9	48h	32 (29.4)	0.59 (0.25-1.35)	NA	7
Wu et al, 2015; China [53]	No/no/yes	Hospital/482	Nosocomial/overall	Overall	All	28-day crude, 16.8	48h	157 32.6)	0.27 (0.14-0.54)**	0.23 (0.10-0.56)**	8
Hsieh et al, 2015; Taiwan [58]	No/no/yes	ED/246	Community/overall	Liver cirrhosis	All	28-day crude, 23.4	72h	212 (86.2)	3.10 (1.46-6.60)	4.29 (1.65-11.16)	8
Cain et al, 2015; US [61]	Yes/no/yes	Hospital/390	-/overall	Overall	GNB	28-day crude, 13.6	48h	359 (92.1)	2.94 (1.40-5.55)	3.54 (1.67-6.85)	7
Hernandez et al, 2015; Spain [59]	No/yes/yes	Hospital/2605	Community/overall	the elderly,	All	30-day crude, 11.4	24h	2201 (84.5)	1.70 (1.26-2.28)	1.49 (1.03-2.16)	7
Dimopoulos et al, 2015; Greece [60]	Yes/yes/yes	ICU/288	Nosocomial/overall	Overall	All	28-day crude, 33.3	BC result	85 (29.5)	ND	0.48 (0.24-0.95)**	7

Ma et al, 2017; China [38]	No/no/yes	Hospital/168	Nosocomial/overall	Hematology	*E. coli*; ESBL (61.9)	30-day crude, 20.2	24h	129 (76.8)	0.881 (0.3012.581)**	NA	7
Trecarichi et al, 2016; Italy [45]	Yes/yes/yes	Hospital/278	Nosocomial/overall	Hemato-oncology	*K. pneumoniae*; KPC (57.9)	21-day crude, 36.3	0 h	126 (45.3)	5.11 (2.63-9.39)	1.87 (1.08-2.22)	8
Guillamet et al, 2016; US [48]	No/no/yes	Hospital/1031	-/pneumonia	Overall	All	In-hospital crude, 37.1	0 h	844 (81.9)	2.3 (1.7-3.2)	2.3 (1.6-3.2)	7
Yoon et al, 2016; South Korea [43]	Yes/yes/yes	Hospital/345	Healthcare-associated/overall	Overall	*S. aureus*; MRSA (100)	In-hospital crude, 33.0	48h	154 (44.6)	0.99 (0.63-1.56)**	NA	7
Gudiol et al, 2016; Spain [49]	No/no/yes	Hospital/54	-/pneumonia	Hemato-oncology	All	30-day crude, 46.3	BC result	46 (85.2)	ND	NA	6
Savage et al, 2016; Canada [46]	Yes/no/yes	ICU/1190	-/overall	Overall	All	In-hospital crude, 40.0	24h	924 (77.6)	ND	1.02 (0.70-1.48)	8
Yilmaz et al, 2016; Turkey [44]	Yes/yes/yes	Hospital/255	-/overall	Overall	*S. aureus*; MRSA (39.2)	28-day crude, 15.3	0 h	183 (71.8)	ND	NA	7
Calle et al, 2016; Spain [50]	No/no/yes	Hospital/98	-pneumonia	Overall	*S. aureus*; MRSA (42.9)	30-day crude, 46.9	24h	60 (61.2)	ND	NA	6
Abraham et al, 2016; US [52]	No/no/yes	ICU/55	Nosocomial/overall	Overall	All	In-hospital crude. 20.0	24h	43 (78.2)	ND	NA	6
Migiyama et al, 2016; Japan [47]	No/no/yes	Hospital/126	-/overall	Overall	*P. aeruginosa*	30-day crude, 31.7	48h	77 (61.1)	0.31 (0.17-0.59)**	0.21 (0.09-0.51)**	7
Cheng et al, 2016; Taiwan [51]	Yes/no/yes	Hospital/111	-/pneumonia	Overall	*E. coli* and *K. pneumoniae*; ESBL (100)	30-day crude, 40.5	5d	51 (45.9)	0.21 (0.09-0.49)**	0.19 (0.07-0.55)**	8
Yamaga & Shime, 2018; Japan [20]	No/no/yes	ICU/62	-/overall	Overall	All	60-day crude, 48.4	24h	46 (74.2)	0.19 (0.052-0.71)**	0.043 (0.0047-0.23)**	8
Saliba et al, 2018; Spain [21]	No/yes/yes	Hospital/546	Nosocomial/vascular catheter	Overall	All	30-day crude, 13.9	48h	331 (60.6)	ND	NA	6
Lee et al, 2017; Taiwan [40]	No/no/yes	ED/2349	Community/overall	Overall	All	28-day crude, 14.5	48h	1917 (81.6)	2.12 (1.63-2.75)	2.60 (1.88-3.60)	7
Xie et al, 2017; China [31]	No/no/yes	Hospital/852	-/overall	Liver cirrhosis	All	30-day crude, 22.5	24h	612 (71.8)	6.060 (4.282-8.576)	8.278 (5.46-12.55)	7
Li et al, 2017; China [39]	No/no/yes	Hospital/104	-/overall	Overall	*K. pneumoniae*; ESBL (39.4)	30-day crude, 25.0	BC result	79 (76.0)	ND	NA	6
Zhang et al, 2017; China [29]	No/no/yes	Hospital/226	Nosocomial/overall	Overall	Enterococci.; VRE (3.3)	30-day crude, 23.9	24h	213 (94.2)	0.2 (0.1-0.5)**	0.2 (0.1-0.4)**	8
Man et al, 2017; China [37]	No/no/yes	Hospital/853	-/overall	Overall	*K. pneumoniae*; ESBL (12.3)	30-day crude, 20.6	24h	489 (57.3)	ND	1.716 (1.267-2.324)	8
Tagashira et al, 2017; Japan [34]	Yes/no/yes	Hospital/573	-/BTI	Overall	All	30-day crude, 6.6	BC result	440 (76.8)	ND	2.78 (1.27-6.11)	7
Satlin et al, 2017; US [35]	Yes/no/yes	Hospital/121	-/overall	Overall	*E. coli*, *K. pneumoniae*, *Enterobacter* spp.; carbapenem-R (100)	30-day crude, 51.2	48h	55 (45.5)	ND	NA	6
Gradel et al, 2017; Denmark [41]	Yes/no/yes	Population/6483	-/overall	Overall	All	30-day crude, 15.7	BC result	3778* (74.5)	1.18 (1.00-1.02)	0.85 (0.70 - 1.02)	7
Wang X et al, 2017; China [32]	Yes/no/yes	Hospital/40	-/overall	Hematology	A. baumannii; carbapenem-R (32.5)	30-day crude, 32.5	48h	20 (50.0)	5.56 (12.50-25.00)	ND, NS	7
Adrie et al, 2017; France [42]	Yes/yes/yes	ICU/571	Nosocomial/overall	Overall	All	30-day crude, 35.7	24h	418 (73.2)	ND	2.3 (1.8-3.0)	7
Royo-Cebrecos et al, 2017; Spain [36]	No/yes/yes	Hospital/173	-/BTI	Oncology	All	30-day crude, 26.0	0 h	135 (78.0)	ND	NA	6
Wang W et al, 2017; China	No/no/yes	Hospital/138	-/overall	Overall	GNB; MDR (100)	In-hospital crude, 25.4	BC result	91 (65.9)	3.18 (1.44-7.04)	NA	6

[33]											
Yu et al, 2017; Taiwan [30]	No/no/yes	Hospital/49	-/overall	Overall	*K. pneumoniae;* ESBL (100)	In-hospital crude, 55.1	0 h	24 (49.0)	0.42 (0.18-0.95)**	0.88 (0.28-2.77)**	7
Park et al, 2018; South Korea [22]	No/no/yes	Hospital/146	-/pneumonia	Overall	*A. baumannii*; carbapenem-R (100)	28-day crude, 70.6	48h	42 (28.8)	2.3 (1.44-3.67)	3.24 (1.94 - 5.42)	8
Kuo SH, et al, 2018; Taiwan [24]	No/no/yes	ICU/428	Nosocomial/vascular catheter	Overall	All	14-day crude, 32.0	48h	153 (35.7)	ND	1.879 (1.011-3.493)	7
Bartoletti at al, 2018; Europe [28]	Yes/yes/yes	Hospital/312	-/overall	Liver cirrhosis	All	30-day crude, 25.3	24h	190 (60.9)	ND	3.14 (1.93-5.12)	7
Kleinhendler et al, 2018; Israel [25]	No/no/yes	Hospital/180	-/overall	Hemato-oncology	All	30-day crude, 22.8	BC result	65* (82.3)	4.6 (2.2-9.6)	NA	7
Haruki et al, 2018; Japan [26]	No/no/yes	ICU/101	-/overall	Overall	*E. coli;* ESBL (23.8)	28-day crude, 20.8	BC result	87 (86.1)	ND	NA	6
Liu et al, 2018; Taiwan [23]	No/no/yes	Hospital/36	-/overall	Overall	*Citrobacter freundii*	28-day crude, 16.7	48h	25 (69.4)	0.857 (0.132-5.552)	NA	5
Bassetti et al, 2018; Italy [27]	Yes/no/yes	Hospital/337	-/overall	the elderly	*S. aureus*; MRSA (39.2)	30-day crude, 35.7	BC result	70* (60.9)	ND	ND, NS	7
Lim et al, 2019; Australia [18]	No/no/no	Hospital/110	-/overall	Overall	*E. coli* and *K. pneumoniae;* ESBL (88.2)	30-day crude, 20.0	24h	53 (48.2)	ND	NA	6
Chusri et al, 2019; Thailand [19]	No/no/yes	Hospital/68	-/overall	Overall	*A. baumannii*	30-day crude, 41.7	BC result	26* (43.3)	0.12 (0.04-0.37)	0.05 (0.01-0.50)	8
Shargian-Alon et al, 2019; Israel [17]	No/no/yes	Hospital/46	Nosocomial/overall	Hematology	*A. baumannii*	7-day crude, 71.7	48h	22 (47.8)	ND	0.052 (0.005-0.590)**	7
Lee et al, 2019; Taiwan [7]	Yes/no/yes	ED/3194	Community/overall	Overall	All	30-day crude, 14.8	24h	2591 (81.1)	0.51 (0.41-0.64)**	0.62 (0.48–0.79)**	8
Alvarez-Marin et al, 2020; Spain [16]	Yes/yes/yes	Hospital/285	-/overall	Overall	*Enterobacter* spp.; ESBL (5.3)	30-day crude, 19.6	24h	168 (58.9)	ND	NA	6
Seo et al, 2020; South Korea [10]	No/no/yes	Hospital/133	-/overall	Overall	*E. coli* and *K. pneumoniae*; Carbapenem-R (100)	14-day crude, 24.1	48h	54 (40.6)	ND	NA	7
Martinez-Nadal et al, 2020; Spain [15]	Yes/yes/yes	Hospital/1615	-/overall	Hemato-oncology	All	30-day crude, 21.0	0 h	1221 (75.6)	ND	NA	6
Falcone et al, 2020; Italy [8]	Yes/no/yes	ICU/102	Nosocomial/overall	Overall	*K. pneumoniae*; KPC (100)	30-day crude, 45.1	24h	55 (53.9)	ND	NA	6
Babar et al, 2021; Pakistan [14]	No/yes/yes	Hospital/137	-/overall	Overall	GNB: Carbapenem-R (100)	14-day crude, 13.9	0 h	99 (72.3)	4.30 (1.87-9.89)	1.382 (1.132-1.687)	8

EAT = empirical antimicrobial therapy; AOR = adjusted odds ratio; BC = blood culture; CI = confidence interval; ESBL = extended-spectrum beta-lactamase; GC = generation cephalosporin; GNB = Gram-negative bacillus; ICU = intensive care unit; MRSA = methicillin-resistant S. aureus; MDR = multidrug resistant; NA = no analysis; ND = no data; NOQ = Newcastle-Ottawa Quality; NS = no significance; OR = odds ratio; R = resistant; S = significance; VRE = vancomycin-resistant enterococci

* indicated the different patient numbers between the entire study and patients assessed for prognostic effects of EAT.

** indicated the prognostic effect of appropriate EAT.

Supplemental Table 3. Included studies with other outcomes, specific sub-populations, or different EAT definitions.

Study and country	Aimed population, patient No.	Primary outcome, (mortality rate)	Cut-off timeline for EAT (after initial culture)	Patient No. (%) of appropriate EAT	OR (95% C.I.) of inappropriate EAT for mortality	AOR (95% C.I.) of inappropriate EAT for mortality
Byl et al, 1999; Belgium [195]	overall, 428	In-hospital sepsis, 13.0	24h	269, 62.9	ND	0.38 (0.17-0.83)
	septic shock, 47	In-hospital crude, 72.3	24h	36 (76.6)	ND	NA
	septic shock, 47	In-hospital sepsis, 61.7	24h	36 (76.6)	ND	NA
Bouza et al, 2004; Spain [184]	overall, 290	In-hospital sepsis, 17.6	24h	170, 58.6	0.5 (0.3-0.9)*	3.5 (1.5-8.2)
Falagas et al, 2006; Greece [175]	overall, 40	14-day crude, 37.5	72h	18, 45.0	ND	ND, NS
Fang et al, 2006; Taiwan [174]	overall, 162	30-day sepsis, 47.5	48h	43, 26.5	ND	0.99 (0.88-1.12)
Osih et al, 2007; US [163]	overall, 167	In-hospital crude, 36.5	48h	114, 68.2	ND	NA
Evans et al, 2009; US [149]	overall, 235	In-hospital crude, 11.1	48h	147, 62.6	ND	1.05 (0.43-2.55)
Horino et al, 2011; Japan [106]	overall, 134	30-day crude, 20.9	48h	81, 60.4	ND	NA
Schechner et al, 2011; Israel [115]	severe sepsis/septic shock, 22	In-hospital crude, 72.7	24h	14, 63.6	1.8 (1.1-2.8)	NA
Enoch et al, 2011; UK [126]	overall, 203	7-day crude, 12.0	0 h	163, 80.3	6.40 (2.22-18.45)	NA
Lee et al, 2011; South Korea [121]	overall, 164	Early clinical response at D3, 73.8	BC result	135, 82.3	8.79 (3.63-21.27)	11.08 (4.37-28.074)
		Longer LOS (>10day)	BC result	135 (82.3)	2.32 (1.02-5.31)	2.47 (1.04-5.84)
Labelle AJ et al, 2008; US [160]	overall, 245	In-hospital crude, 45.7	48h	100, 40.8	ND	ND, NS
Park et al, 2012; South Korea [98]	overall, 100	14-day crude, 30.0	48h	65, 65.0	ND	NA
		7-day crude, 21.0	48h	65 (65.0)	ND	NA
Retamar et al, 2012; Spain [96]	overall, 801	30-day crude 22.6	BC result	ND	ND	1.56 (1.01-2.40)
Retamar et al, 2013; Spain [5]	overall, 341	14-day crude, 16.4	24h	284, 83.3	2.0 (1.2-3.3)	3.33 (1.42-7.69)
Frakking et al, 2013; Netherland [89]	overall, 232	In-hospital crude, 23	24h	85, 36.6	ND	NA
Gasch et al, 2013; Spain [87]	overall, 579	30-day crude, 32	48h	382, 66.0	1.39 (1.04-1.37)	1.37 (1.02-1.83)
Molina et al, 2013; Spain [80]	overall, 105	30-day crude, 14.3	72h	60, 57.1	ND	NA
Marin et al, 2014; Spain [66]	overall, 528	48-hr crude, 6.8	0 h	408, 77.3	ND	NA
Al-Dorzi et al, 2015; Saudi Arabia [62]	overall, 60	ICU crude, 66.7	BC result	36, 60.0	ND	0.15 (0.03-0.96)*
Yamaga & Shime, 2018; Japan [20]	overall, 62	28-day crude, 38.7	24h	46, 74.2	0.54 (0.28-1.04)*	NA
		In-hospital crude, 45.2	24h	46 (74.2)	0.28 (ND)	NA
Zhang et al, 2017; China [29]	overall, 226	7-day crude, 11.1	24h	213, 94.2	0.0 (0.0-0.1)*	0.2 (0.0-0.7)*
Satlin et al, 2017; US [35]	overall, 121	30-day crude, 51.2	12h	19, 15.7	ND	NA
Gradel et al, 2017; Denmark [41]	overall, 5068	1-year crude, 20.6	BC result	3778, 74.5	1.54 (1.32-1.80)	1.35 (1.13-1.60)
Park et al, 2018; South Korea [22]	overall, 146	14-day crude, 65.8	5 day	42, 28.8	ND	NA
Kleinhendler et al, 2018; Israel [25]	overall, 79	7-day crude, 4.6	BC result	65, 82.3	9.3 (2.63-32.7)	NA
Bassetti et al, 2018; Italy[27]	overall, 115	7-day crude, 20.3	BC result	72, 62.6	ND	0.3 (0.1-0.9)*
Shargian-Alon et al, 2019; Israel [17]	overall, 46	7-day crude, 71.7	24h	11, 23.9	ND	NA
Lee et al, 2019; Taiwan [7]	overall, 3194	30-day crude, 14.8	48h	2719, 85.1	0.434 (0.343-0.548)*	0.54 (0.43–0.71)*
	critically ill, 646	30-day crude, 46.7	24h	526 (81.4)	0.564 (0.377-0.842)*	0.50 (0.33–0.78)*
	critically ill, 646	30-day crude, 46.7	48h	546 (84.5)	0.505 (0.327-0.781)*	0.42 (0.27–0.67)*
Martinez-Nadal et al, 2020; Spain [15]	*Pseudomonas* spp., 251	30-day crude, 34.7	0 h	201, 80.1	ND	3.02 (1.29-7.07)

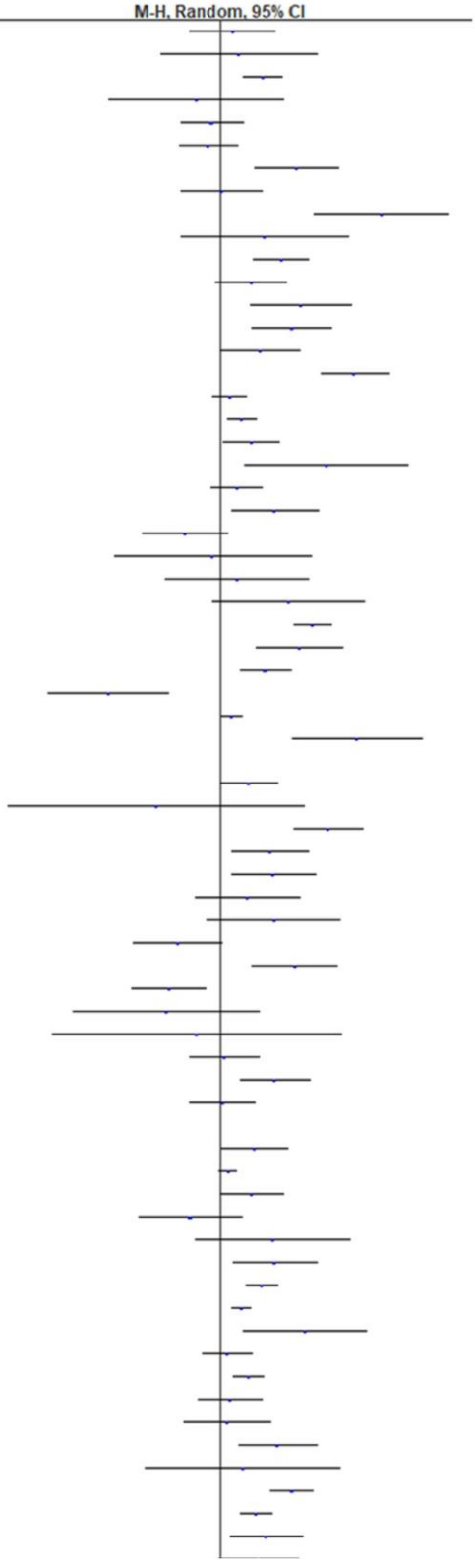
EAT = empirical antimicrobial therapy;; AOR = adjusted odds ratio; BC = blood culture; CI = confidence interval; ICU = intensive care unit; LOS = length of hospital stay; OR = odds ratio; NA = no analysis; ND =no data; NS =no significance.

* indicated the prognostic effect of appropriate EAT.

Supplemental Figure 1. Unadjusted analyses in overall patients: the forest (1A) and funnel plots (1B).

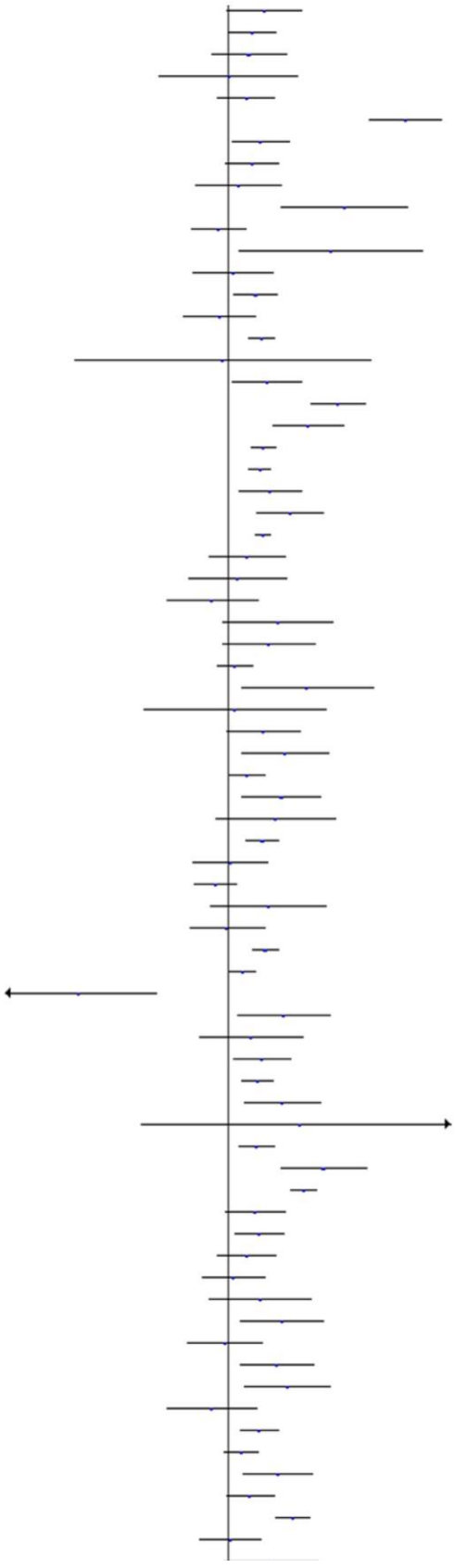
(1A) Forest plots

Study or Subgroup	Inappropriate Events	Inappropriate Total	Appropriate Events	Appropriate Total	Weight	Odds Ratio M-H, Random, 95% CI
Abhilash 2010	13	47	17	74	0.5%	1.28 [0.55, 2.96]
Abraham 2016	3	12	8	43	0.2%	1.46 [0.32, 6.64]
Adrie 2017	77	153	127	418	0.7%	2.32 [1.59, 3.39]
Al-Dorzi 2015	21	24	33	36	0.2%	0.64 [0.12, 3.45]
Alvarez-Marin 2020	21	117	34	168	0.6%	0.86 [0.47, 1.58]
Ammerlaan 2009	20	94	60	240	0.6%	0.81 [0.46, 1.44]
Anatoliotaki 2004	18	46	14	111	0.5%	4.45 [1.97, 10.06]
Asgeirsson 2011	9	63	36	262	0.5%	1.05 [0.48, 2.30]
Babar 2021	16	38	3	99	0.3%	23.27 [6.23, 86.88]
Bang 2013	10	43	2	18	0.2%	2.42 [0.47, 12.39]
Bartoletti 2014	48	122	31	190	0.6%	3.33 [1.96, 5.65]
Bartoletti 2018	23	62	24	100	0.6%	1.87 [0.94, 3.72]
Bassetti 2007	25	31	40	87	0.4%	4.90 [1.83, 13.12]
Bassetti 2012	21	56	14	109	0.5%	4.07 [1.87, 8.88]
Bassetti 2018	21	45	20	70	0.5%	2.19 [1.00, 4.78]
Bodey 1986	50	62	129	559	0.6%	13.89 [7.18, 26.87]
Bodro 2014	66	268	187	880	0.7%	1.21 [0.88, 1.67]
Boel 2015	79	306	313	1702	0.8%	1.54 [1.16, 2.05]
Bouza 2004	37	120	33	170	0.6%	1.85 [1.08, 3.18]
Bouza 2005	10	12	25	65	0.2%	8.00 [1.62, 39.56]
Byl 1999	37	159	48	269	0.7%	1.40 [0.86, 2.26]
Cain 2015	9	31	44	359	0.5%	2.93 [1.27, 6.77]
Calle 2016	14	38	32	60	0.5%	0.51 [0.22, 1.17]
Carratala 1998	3	5	22	35	0.2%	0.89 [0.13, 6.02]
Chamot 2003	3	17	13	98	0.3%	1.40 [0.35, 5.55]
Chang 2009	6	33	3	55	0.3%	3.85 [0.89, 16.62]
Chen 2013	97	255	62	682	0.7%	6.14 [4.27, 8.83]
Cheng 2016	34	60	11	51	0.5%	4.76 [2.05, 11.02]
Cheong 2008	29	106	68	508	0.7%	2.44 [1.48, 4.01]
Chusri 2019	7	34	18	26	0.3%	0.12 [0.04, 0.37]
Corona 2010	415	1013	244	689	0.8%	1.27 [1.04, 1.55]
Daikos 2009	9	29	4	133	0.3%	14.51 [4.08, 51.60]
Davis 2014	0	0	21	41		Not estimable
Dimopoulos 2015	75	203	21	85	0.6%	1.79 [1.01, 3.16]
Elhanan 1997	0	10	17	124	0.1%	0.29 [0.02, 5.22]
Endimiani 2003	48	207	11	314	0.6%	8.32 [4.20, 16.46]
Enoch 2011	15	40	30	163	0.5%	2.66 [1.25, 5.65]
Erbay 2009	39	60	17	43	0.5%	2.84 [1.26, 6.38]
Evans 2009	8	88	8	147	0.4%	1.74 [0.63, 4.81]
Falagas 2006	13	22	6	18	0.3%	2.89 [0.79, 10.57]
Falcone 2014	15	52	20	42	0.5%	0.45 [0.19, 1.05]
Falcone 2020	30	47	16	55	0.5%	4.30 [1.87, 9.89]
Fang 2006	36	119	23	43	0.5%	0.38 [0.18, 0.77]
Feldman 1990	2	6	24	41	0.2%	0.35 [0.06, 2.16]
Feodoroff 2011	1	46	1	30	0.1%	0.64 [0.04, 10.71]
Frakking 2013	30	147	16	85	0.6%	1.11 [0.56, 2.17]
Garey 2006	24	62	23	130	0.6%	2.94 [1.49, 5.81]
Garnacho-Montero 2013	44	122	23	66	0.6%	1.05 [0.56, 1.97]
Gasch 2013	312	201	18	378		Not estimable
Girometti 2014	40	124	18	93	0.6%	1.98 [1.05, 3.75]
Gradel 2017	224	1290	571	3778	0.8%	1.18 [1.00, 1.40]
Gudiol 2011	20	59	62	290	0.6%	1.89 [1.03, 3.46]
Gudiol 2013	7	33	19	59	0.4%	0.57 [0.21, 1.54]
Gudiol 2016	6	9	19	46	0.3%	2.84 [0.63, 12.80]
Guilarde 2006	22	44	17	67	0.5%	2.94 [1.31, 6.59]
Guillamet 2016	100	187	282	844	0.7%	2.29 [1.66, 3.16]
Hanon 2002	279	768	351	1290	0.8%	1.53 [1.26, 1.85]
Haruki 2018	7	14	14	87	0.3%	5.21 [1.58, 17.20]
Hernandez 2014	28	189	72	556	0.7%	1.17 [0.73, 1.87]
Hernandez 2015	69	404	231	2201	0.8%	1.76 [1.31, 2.35]
Horcajada 2012	15	127	53	540	0.6%	1.23 [0.67, 2.26]
Horino 2011	14	63	14	71	0.5%	1.16 [0.51, 2.68]
Hsieh 2015	15	34	43	212	0.5%	3.10 [1.46, 6.60]
Hung 2005	2	9	4	26	0.2%	1.57 [0.24, 10.49]
Ibrahim 2000	91	147	98	345	0.7%	4.10 [2.73, 6.15]
Johnson 2011	125	235	185	519	0.7%	2.05 [1.50, 2.81]
Jung 2012	13	40	84	513	0.6%	2.46 [1.22, 4.96]



(1A) Forest plots (continued)

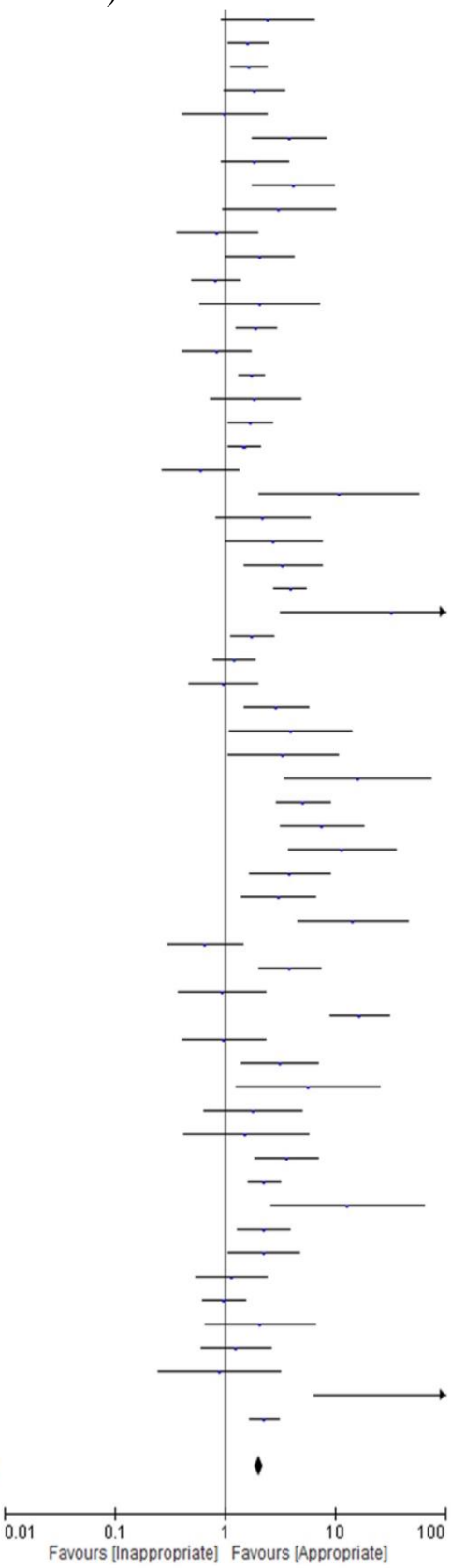
Kang 2003	40	89	13	47	0.5%	2.14 [0.99, 4.58]
Kang 2005	58	151	37	135	0.7%	1.65 [1.00, 2.72]
Kang 2011	10	64	32	301	0.5%	1.56 [0.72, 3.35]
Kang 2013	7	64	3	28	0.3%	1.02 [0.24, 4.28]
Kang Wi 2013	27	257	22	299	0.6%	1.48 [0.82, 2.66]
Khan 2010	55	65	47	387	0.5%	39.79 [18.99, 83.36]
Khatib 2006	21	60	60	282	0.6%	1.99 [1.09, 3.64]
Kim 2006	46	117	34	121	0.6%	1.66 [0.96, 2.85]
Kim 2014	7	38	45	296	0.5%	1.26 [0.52, 3.03]
Kleinhendler 2018	9	14	9	65	0.3%	11.20 [3.05, 41.10]
Klevay 2009	21	78	75	244	0.6%	0.83 [0.47, 1.47]
Ku 2011	6	8	5	19	0.2%	8.40 [1.26, 56.07]
Kuo 2012	13	131	12	135	0.5%	1.13 [0.50, 2.57]
Kuo SH 2018	100	275	37	153	0.7%	1.79 [1.15, 2.79]
Labelle A 2012	60	208	12	37	0.5%	0.84 [0.40, 1.79]
Labelle AJ 2012	321	470	224	436	0.8%	2.04 [1.56, 2.67]
Lee 2011	0	29	2	135	0.1%	0.91 [0.04, 19.35]
Lee 2013	32	63	21	67	0.5%	2.26 [1.11, 4.62]
Lee 2014	75	121	25	177	0.6%	9.91 [5.66, 17.36]
Lee 2015	24	41	31	148	0.5%	5.33 [2.55, 11.13]
Lee 2017	100	432	239	1917	0.8%	2.11 [1.63, 2.75]
Lee 2019	136	603	336	2591	0.8%	1.95 [1.56, 2.44]
Lee CC 2012	19	124	23	330	0.6%	2.42 [1.26, 4.61]
Lee NY 2012	63	177	12	91	0.6%	3.64 [1.84, 7.19]
Leibovici 1998	432	1255	436	2158	0.8%	2.07 [1.77, 2.43]
Lewis 2011	27	112	11	63	0.5%	1.50 [0.69, 3.28]
Li 2017	7	25	19	79	0.4%	1.23 [0.45, 3.39]
Lim 2019	10	57	12	53	0.4%	0.73 [0.28, 1.86]
Lin 2004	7	16	15	70	0.4%	2.85 [0.91, 8.92]
Lin 2006	12	41	10	67	0.4%	2.36 [0.91, 6.10]
Lin 2008	50	540	79	983	0.7%	1.17 [0.81, 1.69]
Lin 2010	17	23	6	17	0.3%	5.19 [1.33, 20.28]
Liu 2018	2	11	4	25	0.2%	1.17 [0.18, 7.56]
Lodise 2003	16	48	23	119	0.5%	2.09 [0.98, 4.43]
Lodise 2007	21	48	10	52	0.5%	3.27 [1.33, 7.99]
Lye 2012	77	294	73	381	0.7%	1.50 [1.04, 2.16]
Ma 2017	14	39	20	129	0.5%	3.05 [1.36, 6.86]
Mackenzie 2003	6	21	7	54	0.3%	2.69 [0.78, 9.24]
Man 2017	100	364	76	489	0.7%	2.06 [1.47, 2.88]
Marcos 2008	10	82	27	232	0.5%	1.05 [0.49, 2.29]
Marin 2014	123	414	40	114	0.7%	0.78 [0.50, 1.21]
Marra 2006	6	27	7	64	0.3%	2.33 [0.70, 7.72]
Marschall 2008	11	79	24	171	0.5%	0.99 [0.46, 2.14]
Martinez 2010	79	454	388	4409	0.8%	2.18 [1.68, 2.84]
Martinez-Nadal 2020	99	394	240	1221	0.8%	1.37 [1.05, 1.79]
Metan 2005	5	41	9	12	0.2%	0.05 [0.01, 0.23]
Metan 2009	53	76	10	24	0.4%	3.23 [1.25, 8.32]
Meyers 1989	8	16	32	84	0.4%	1.63 [0.55, 4.76]
Micek & Lioyd 2010	23	75	41	230	0.6%	2.04 [1.12, 3.70]
Micek 2010	270	522	87	238	0.7%	1.86 [1.36, 2.55]
Migiyama 2016	23	49	17	77	0.5%	3.12 [1.43, 6.80]
Molina 2013	1	13	0	18	0.1%	4.44 [0.17, 118.00]
Morata 2012	65	246	76	463	0.7%	1.83 [1.26, 2.66]
Ortega 2007	14	38	12	162	0.5%	7.29 [3.01, 17.64]
Ortega 2009	100	349	340	4409	0.8%	4.81 [3.72, 6.22]
Ortega 2011	14	86	82	824	0.6%	1.76 [0.95, 3.26]
Ortega 2012	20	131	106	1242	0.6%	1.93 [1.15, 3.24]
Ortega 2013	16	155	46	643	0.6%	1.49 [0.82, 2.72]
Osih 2007	26	68	35	99	0.6%	1.13 [0.60, 2.15]
Park 2011	7	42	10	108	0.4%	1.96 [0.69, 5.55]
Park 2012	24	35	27	65	0.5%	3.07 [1.29, 7.31]
Park 2013	17	76	17	73	0.5%	0.95 [0.44, 2.04]
Park 2018	80	104	23	42	0.5%	2.75 [1.29, 5.89]
Park H 2015	21	48	10	54	0.5%	3.42 [1.40, 8.35]
Park SY 2015	17	77	9	32	0.4%	0.72 [0.28, 1.85]
Paul 2010	168	342	56	168	0.7%	1.93 [1.31, 2.84]
Pena 2013	86	261	90	332	0.7%	1.32 [0.93, 1.88]
Peralta 2007	12	106	24	557	0.5%	2.84 [1.37, 5.86]
Peralta 2012	47	189	34	198	0.7%	1.60 [0.97, 2.62]
Phillips 1990	67	159	215	1346	0.7%	3.83 [2.71, 5.42]
Picot-Gueraud 2015	21	91	30	137	0.6%	1.07 [0.57, 2.02]



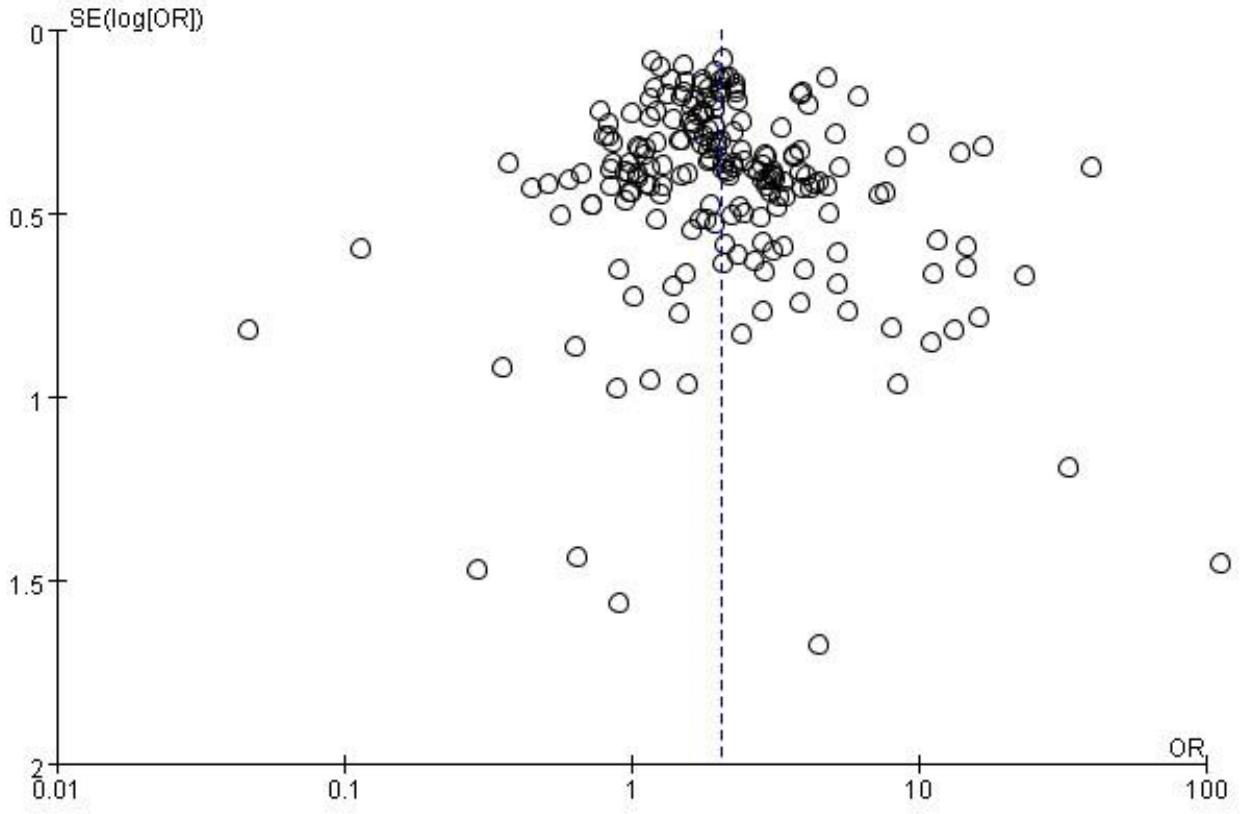
(1A) Forest plots (continued)

Rebelo 2011	24	89	6	46	0.4%	2.46 [0.93, 6.54]
Reisfeld 2011	70	151	79	227	0.7%	1.62 [1.06, 2.47]
Retamar 2012	49	199	99	602	0.7%	1.66 [1.13, 2.45]
Retamar 2013	17	57	53	284	0.6%	1.85 [0.98, 3.52]
Rodriguez-Bano 2010	13	43	16	53	0.5%	1.00 [0.42, 2.41]
Rodriguez-Bano J 2010	17	77	14	205	0.5%	3.87 [1.80, 8.30]
Rodriguez-Pardo 2014	22	134	15	157	0.6%	1.86 [0.92, 3.75]
Rong 2012	21	46	12	72	0.5%	4.20 [1.80, 9.82]
Rosa 2011	6	17	11	74	0.3%	3.12 [0.96, 10.20]
Royo-Cebrecos 2017	9	38	36	135	0.5%	0.85 [0.37, 1.98]
Ruiz-Giardin 2013	14	71	24	226	0.5%	2.07 [1.00, 4.25]
Saliba 2017	27	215	49	331	0.7%	0.83 [0.50, 1.37]
Salonen 1997	9	29	5	28	0.3%	2.07 [0.60, 7.20]
Sancho 2012	106	155	114	216	0.7%	1.94 [1.26, 2.98]
Satlin 2017	31	66	28	55	0.5%	0.85 [0.42, 1.75]
Savage 2016	135	266	341	924	0.8%	1.76 [1.34, 2.32]
Schechner 2011	15	32	14	44	0.4%	1.89 [0.74, 4.84]
Schramm 2006	99	390	28	169	0.7%	1.71 [1.08, 2.73]
Schweizer 2010	77	277	109	537	0.7%	1.51 [1.08, 2.12]
Seo 2020	16	79	16	54	0.5%	0.60 [0.27, 1.34]
Shargian-Alon 2019	22	24	11	22	0.2%	11.00 [2.07, 58.52]
Shih 2005	20	41	9	30	0.4%	2.22 [0.82, 5.99]
Shime 2011	6	40	16	270	0.4%	2.80 [1.03, 7.65]
Shorr 2014	54	93	11	38	0.5%	3.40 [1.51, 7.66]
Son 2020	77	214	116	930	0.7%	3.94 [2.81, 5.54]
Song 2011	11	15	1	13	0.1%	33.00 [3.18, 342.26]
Soriano 2000	33	189	77	719	0.7%	1.76 [1.13, 2.75]
Soriano 2008	73	246	43	168	0.7%	1.23 [0.79, 1.91]
Su 2007	23	57	27	66	0.5%	0.98 [0.47, 2.01]
Tagashira 2017	17	133	21	440	0.6%	2.92 [1.49, 5.72]
Takesue 2011	41	106	3	22	0.3%	3.99 [1.11, 14.35]
Tam 2010	6	16	14	93	0.3%	3.39 [1.06, 10.81]
Trecarichi 2009	7	11	5	51	0.2%	16.10 [3.46, 74.84]
Trecarichi 2016	79	152	22	126	0.6%	5.12 [2.92, 8.95]
Tumbarello 2008	29	56	9	73	0.5%	7.64 [3.19, 18.28]
Tumbarello 2010	11	27	6	107	0.4%	11.57 [3.75, 35.68]
Tumbarello 2011	20	37	16	69	0.5%	3.90 [1.66, 9.16]
Tumbarello 2012	39	75	13	50	0.5%	3.08 [1.42, 6.71]
Tumbarello 2013	16	21	14	78	0.4%	14.63 [4.59, 46.61]
Tuon 2011	20	49	28	55	0.5%	0.67 [0.31, 1.45]
Valles 2003	34	49	107	290	0.6%	3.88 [2.02, 7.45]
Vitkauskiene 2010	18	31	29	49	0.4%	0.95 [0.38, 2.38]
Wang 2005	50	68	43	303	0.6%	16.80 [8.96, 31.47]
Wang 2011	15	63	12	50	0.5%	0.99 [0.41, 2.36]
Wang W 2017	19	47	16	91	0.5%	3.18 [1.44, 7.04]
Wang X 2017	10	20	3	20	0.3%	5.67 [1.25, 25.61]
Wareham 2008	10	74	7	88	0.4%	1.81 [0.65, 5.01]
Wu 2012	9	38	4	24	0.3%	1.55 [0.42, 5.74]
Wu 2015	70	325	11	157	0.6%	3.64 [1.87, 7.10]
Xie 2017	81	240	111	612	0.7%	2.30 [1.64, 3.22]
Yamaga & Shime 2017	14	16	16	46	0.2%	13.13 [2.65, 65.07]
Yang 2013	72	131	32	91	0.6%	2.25 [1.30, 3.90]
Ye 2006	33	68	17	58	0.5%	2.27 [1.09, 4.76]
Yilmaz 2016	12	72	27	183	0.5%	1.16 [0.55, 2.43]
Yoon 2016	63	191	51	154	0.7%	0.99 [0.63, 1.56]
Yu 2017	16	25	11	24	0.4%	2.10 [0.67, 6.60]
Zaragoza 2003	22	39	64	127	0.5%	1.27 [0.62, 2.62]
Zarkotou 2011	13	39	5	14	0.3%	0.90 [0.25, 3.24]
Zhang 2017	13	13	41	213	0.1%	112.23 [6.54, 1926.54]
Ziberberg 2014	106	245	205	819	0.8%	2.28 [1.70, 3.08]
Total (95% CI)		**25954**		**64008**	**100.0%**	**2.06 [1.88, 2.25]**
Total events	8662		10815			

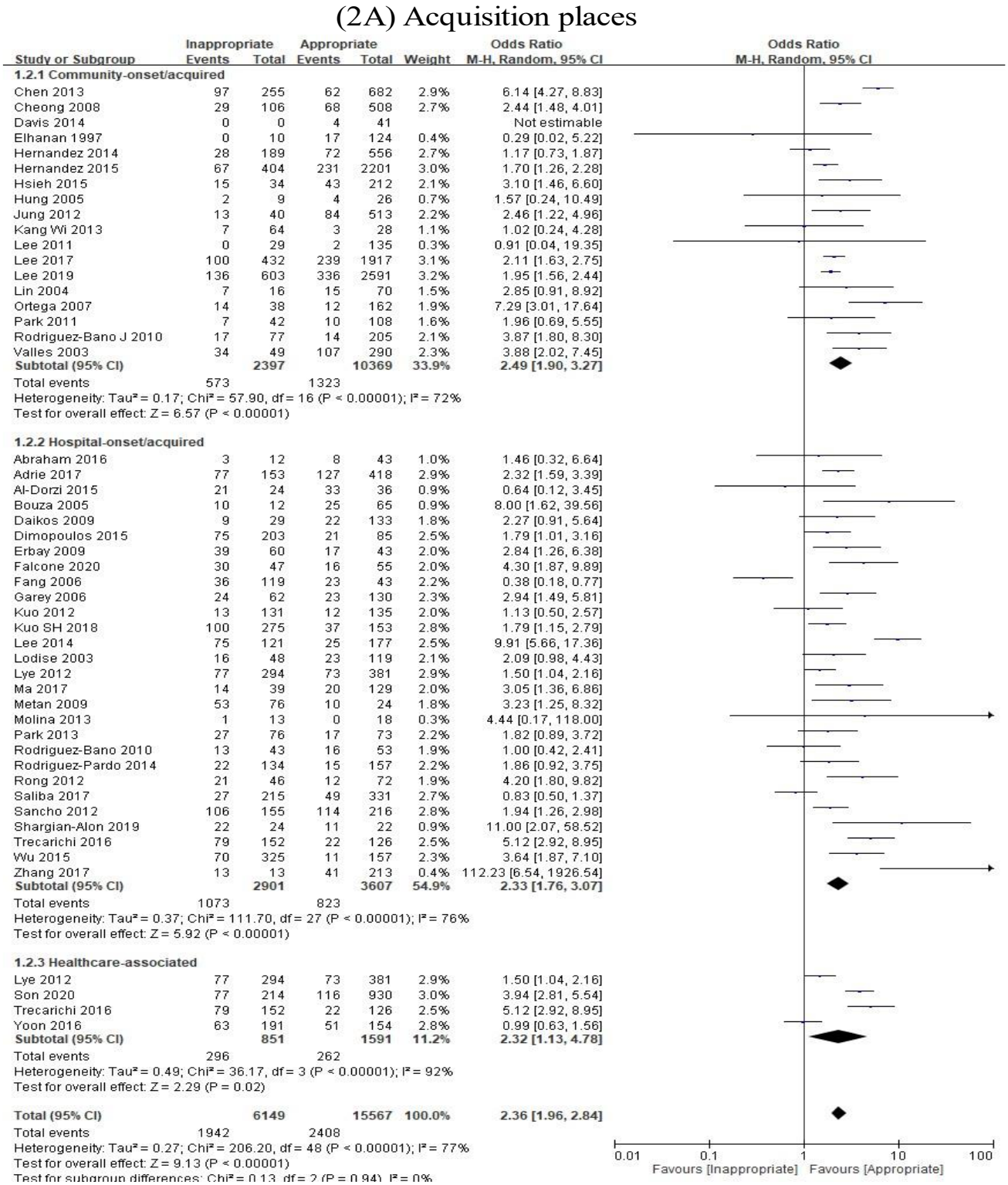
Heterogeneity: Tau² = 0.26; Chi² = 885.51, df = 195 (P < 0.00001); I² = 78%
Test for overall effect: Z = 15.63 (P < 0.00001)



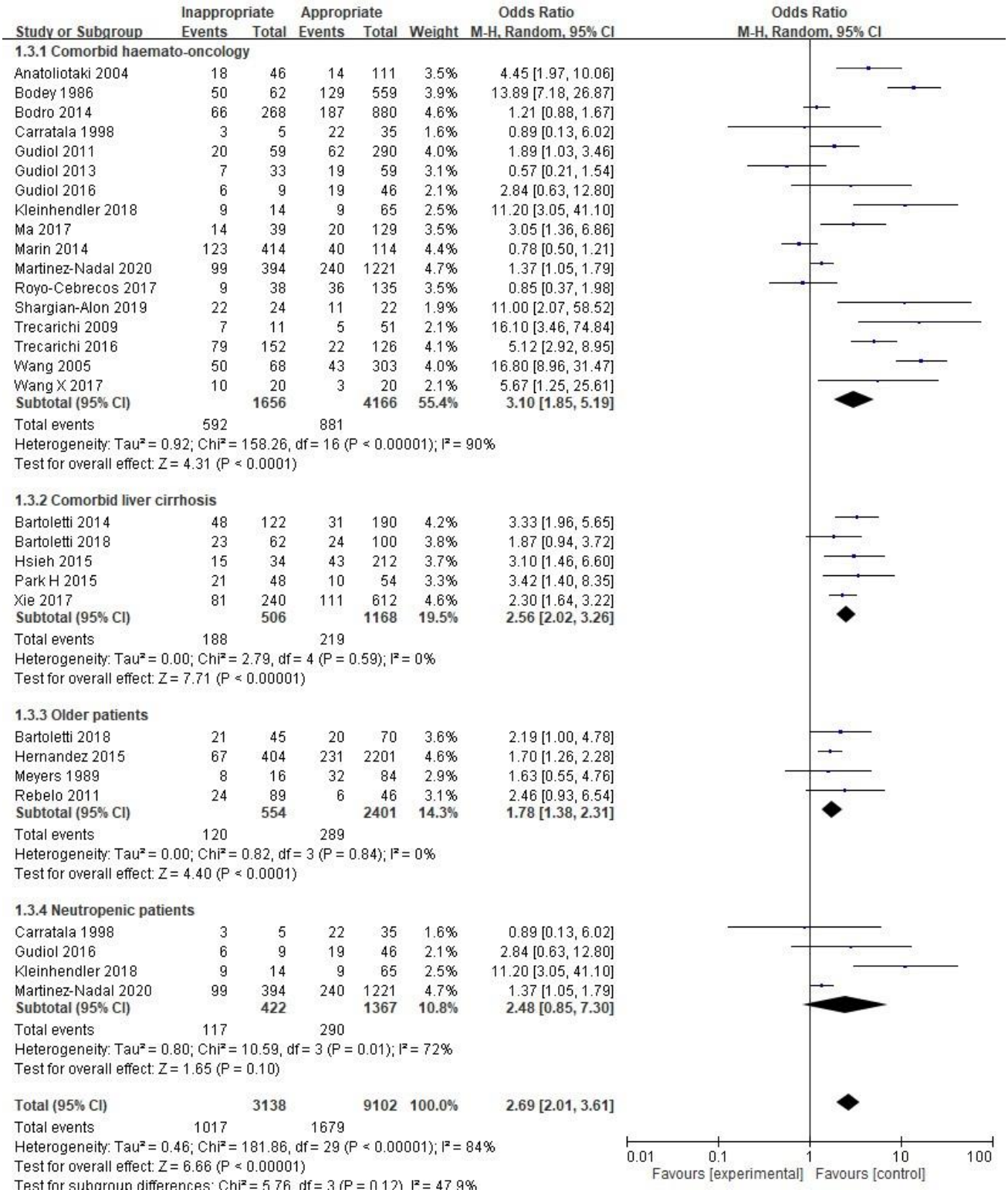
(1B) Funnel plots



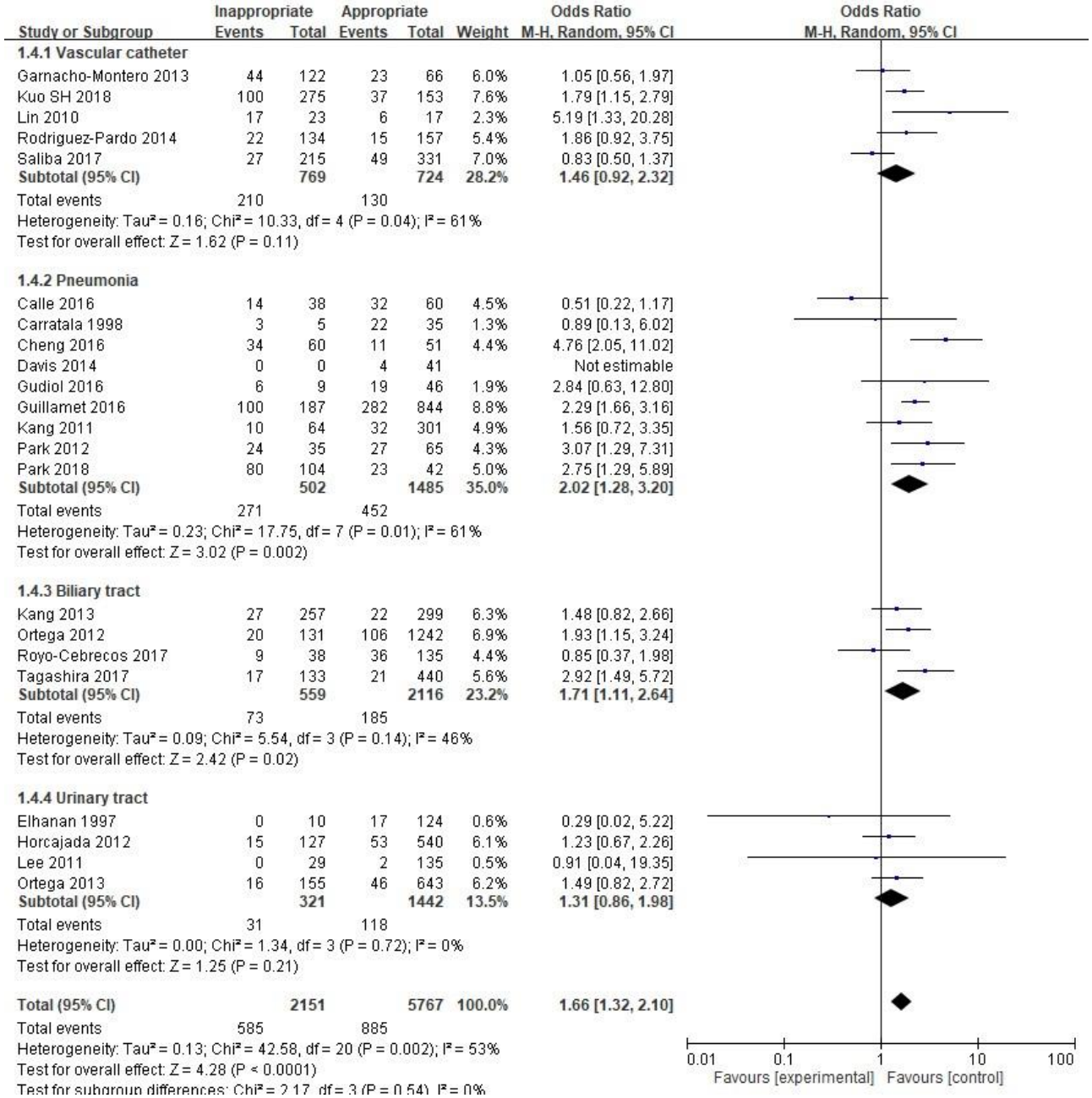
Supplemental Figure 2. Unadjusted analyses in subgroup patients: (2A) acquisition places; (2B) specific comorbidities or conditions; (2C) bacteraemia sources; (2D) bacteraemia severity; (2E) *Staphylococcus aureus*; (2F) Enterobacteriaceae, by acquisition places; (2G) Enterobacteriaceae, by species; (2H) glucose non-fermentative rods; (2I) antibiotic-resistant microorganisms; (2J) varied timeliness of empirical antimicrobial therapy (EAT); and (2K) varied mortality timeliness



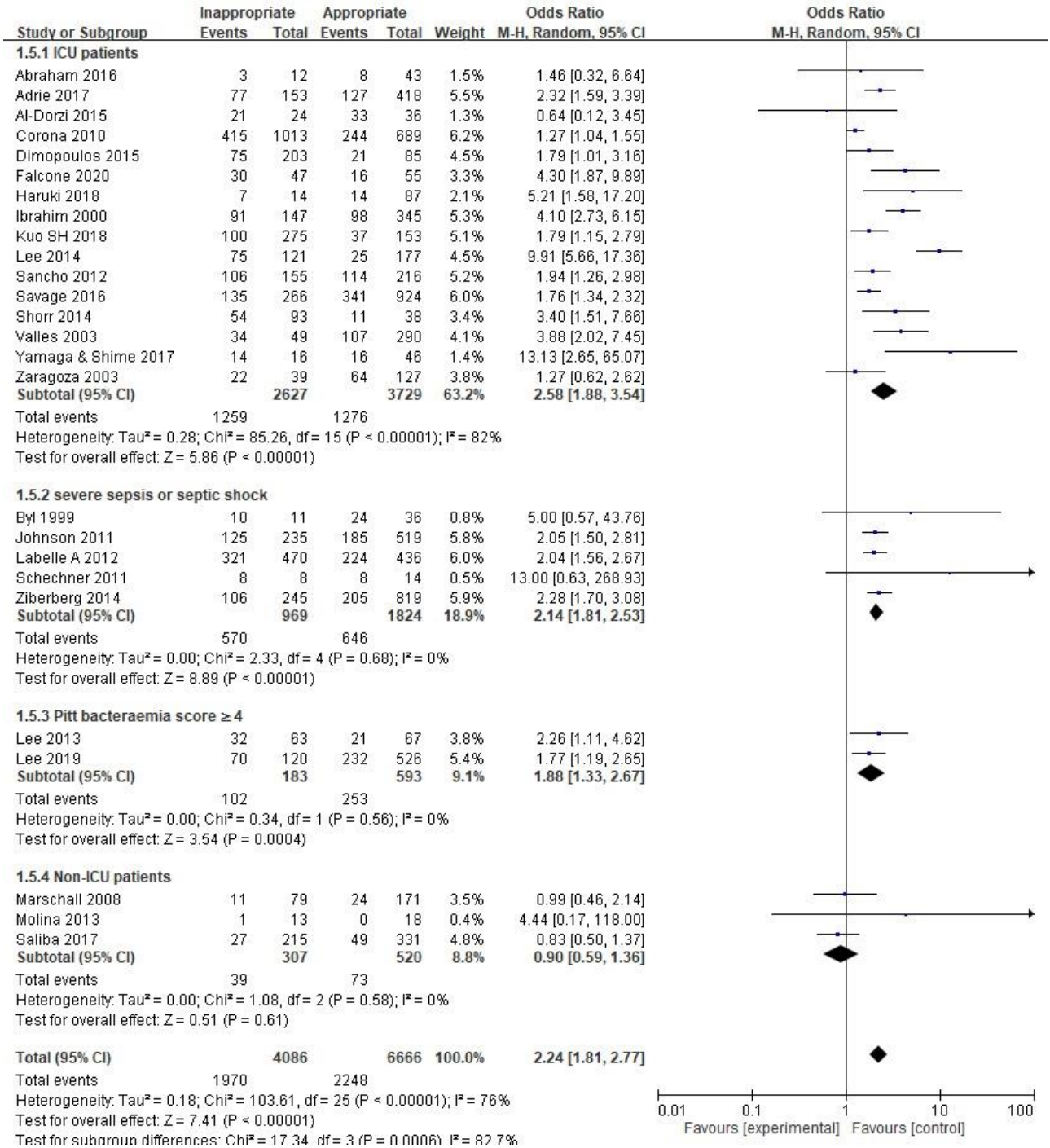
(2B) Specific comorbidity or condition



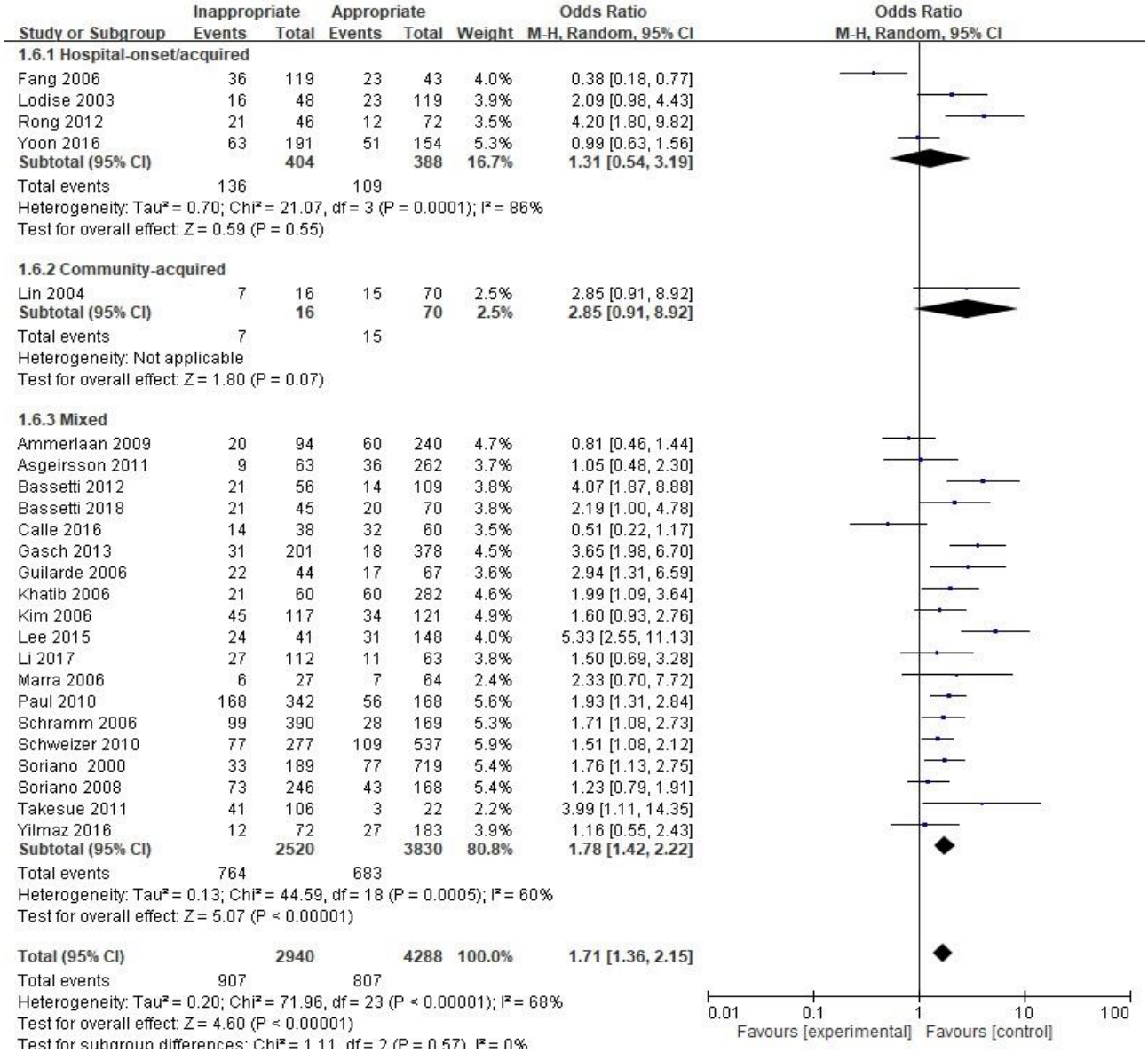
(2C) Bacteraemia source



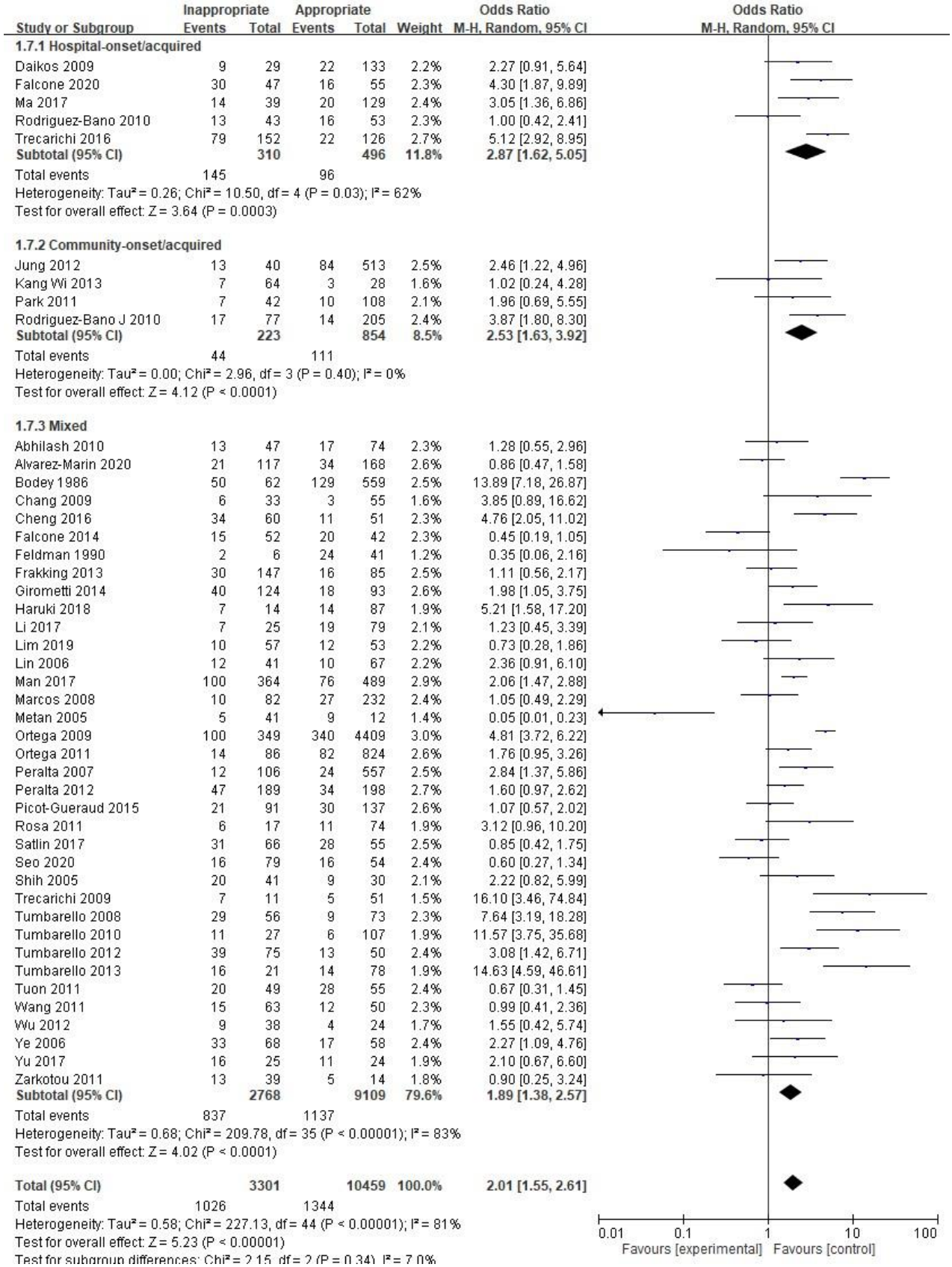
(2D) Bacteraemia severity



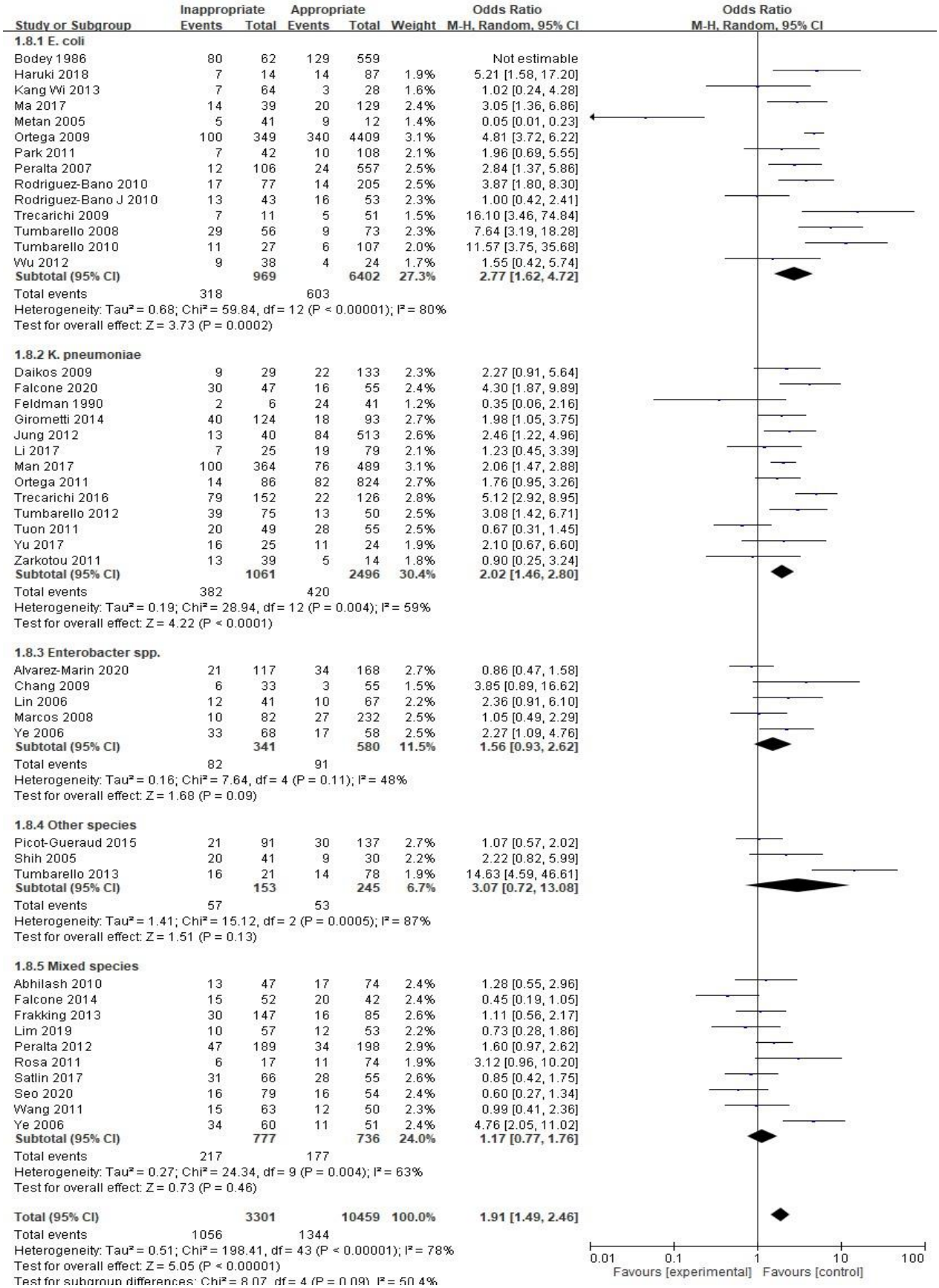
(2E) Staphylococcus aureus



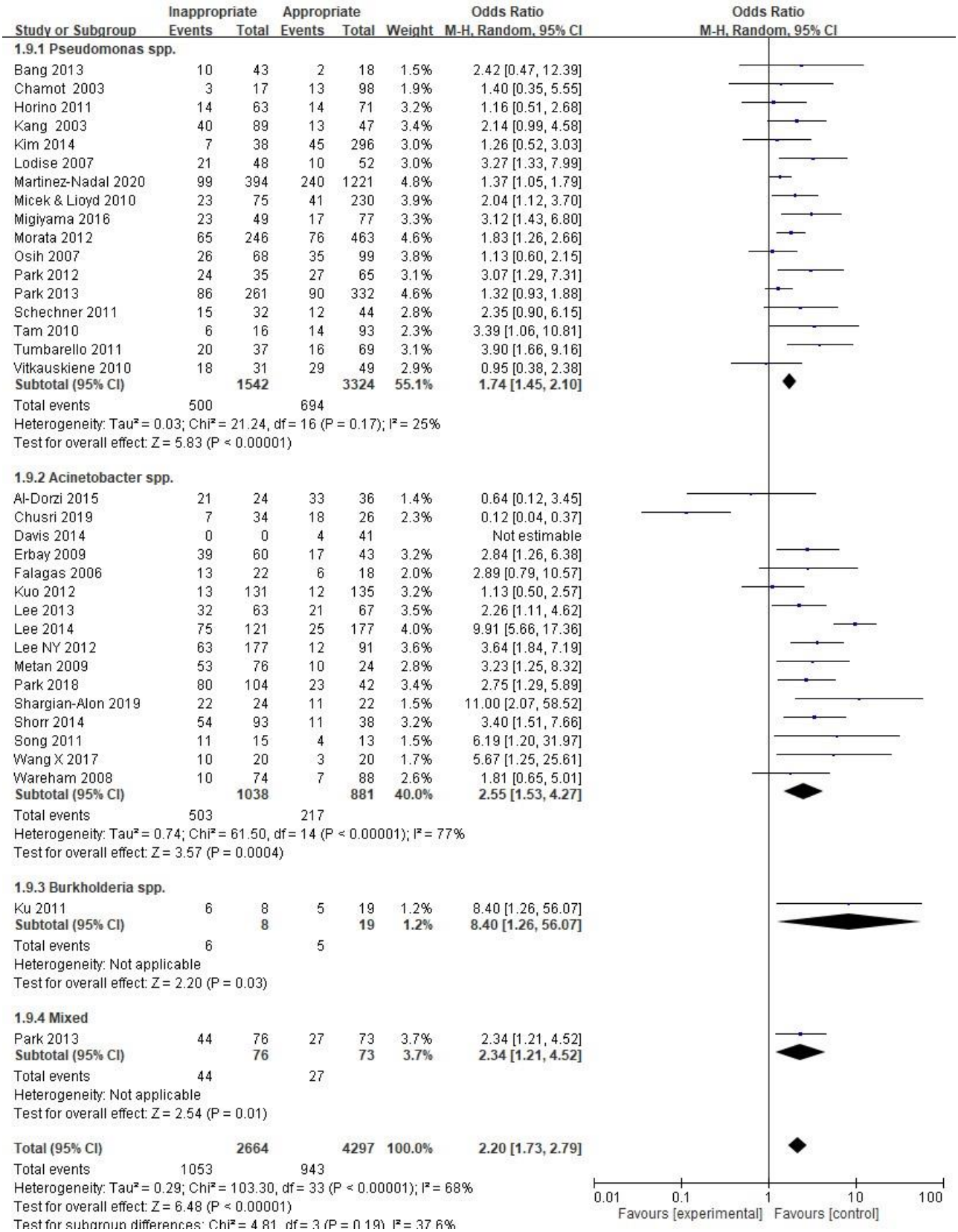
(2F) Enterobacteriaceae, by acquisition places places



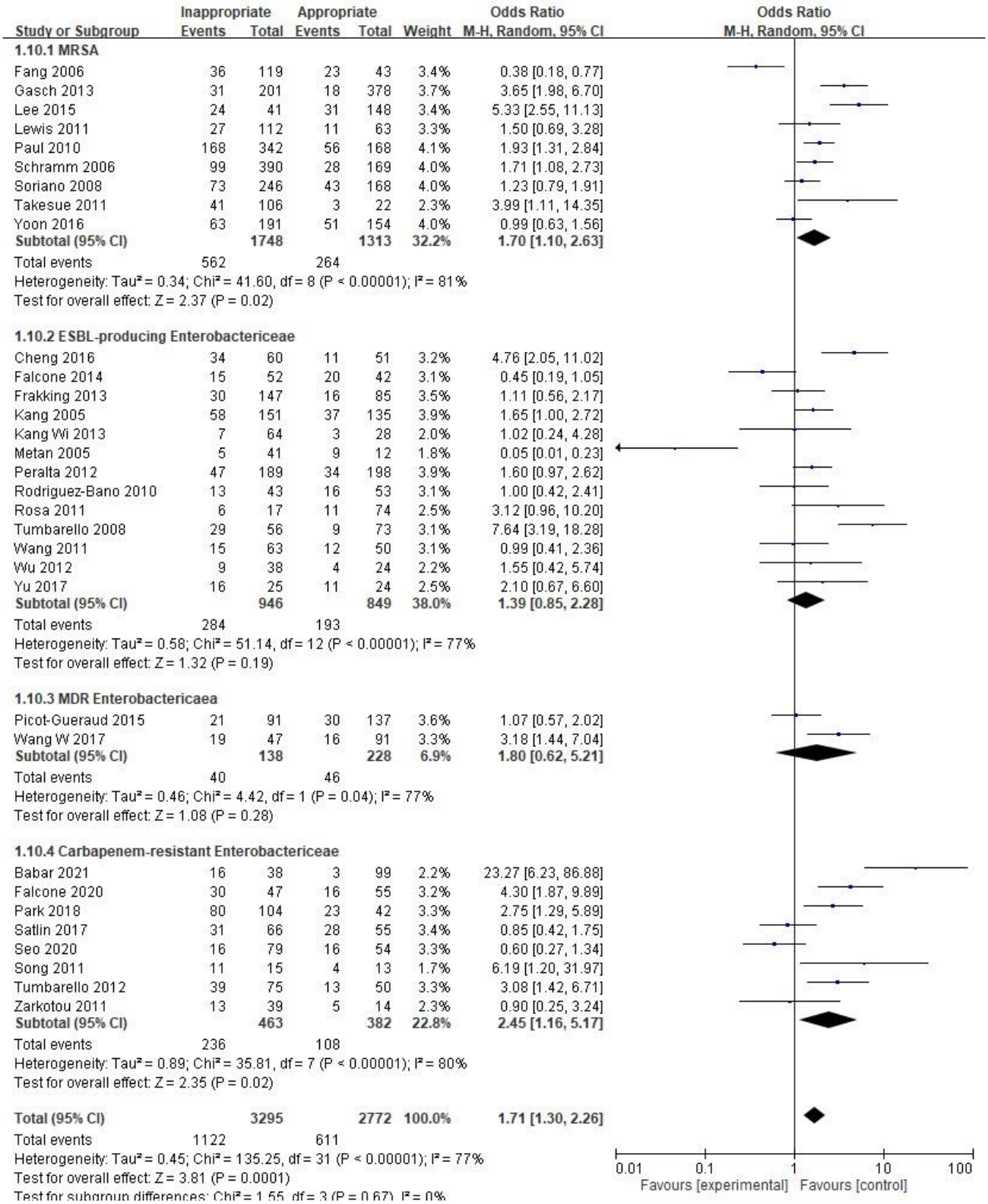
(2G) Enterobacteriaceae, by species



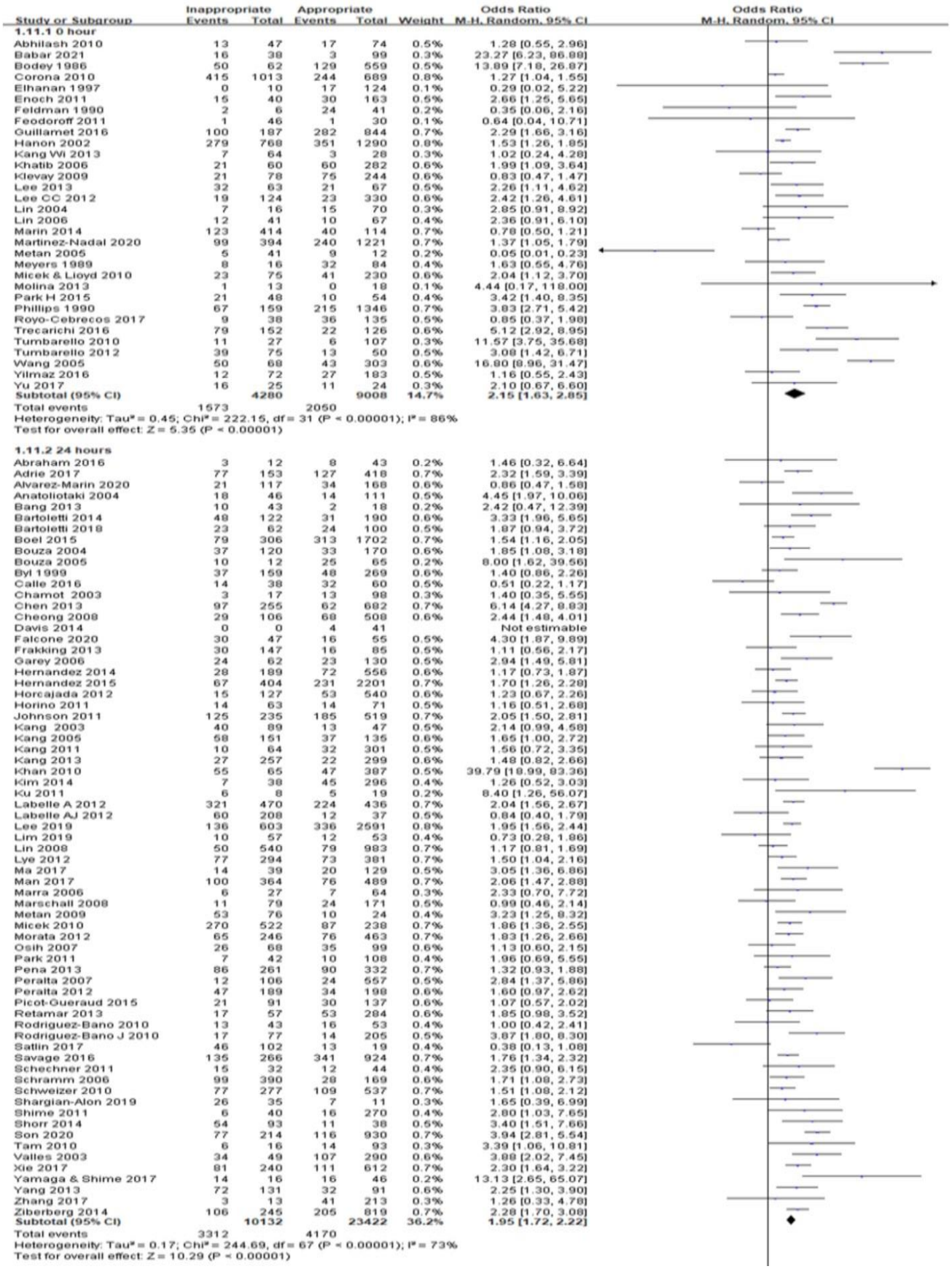
(2H) Glucose non-fermentative rods



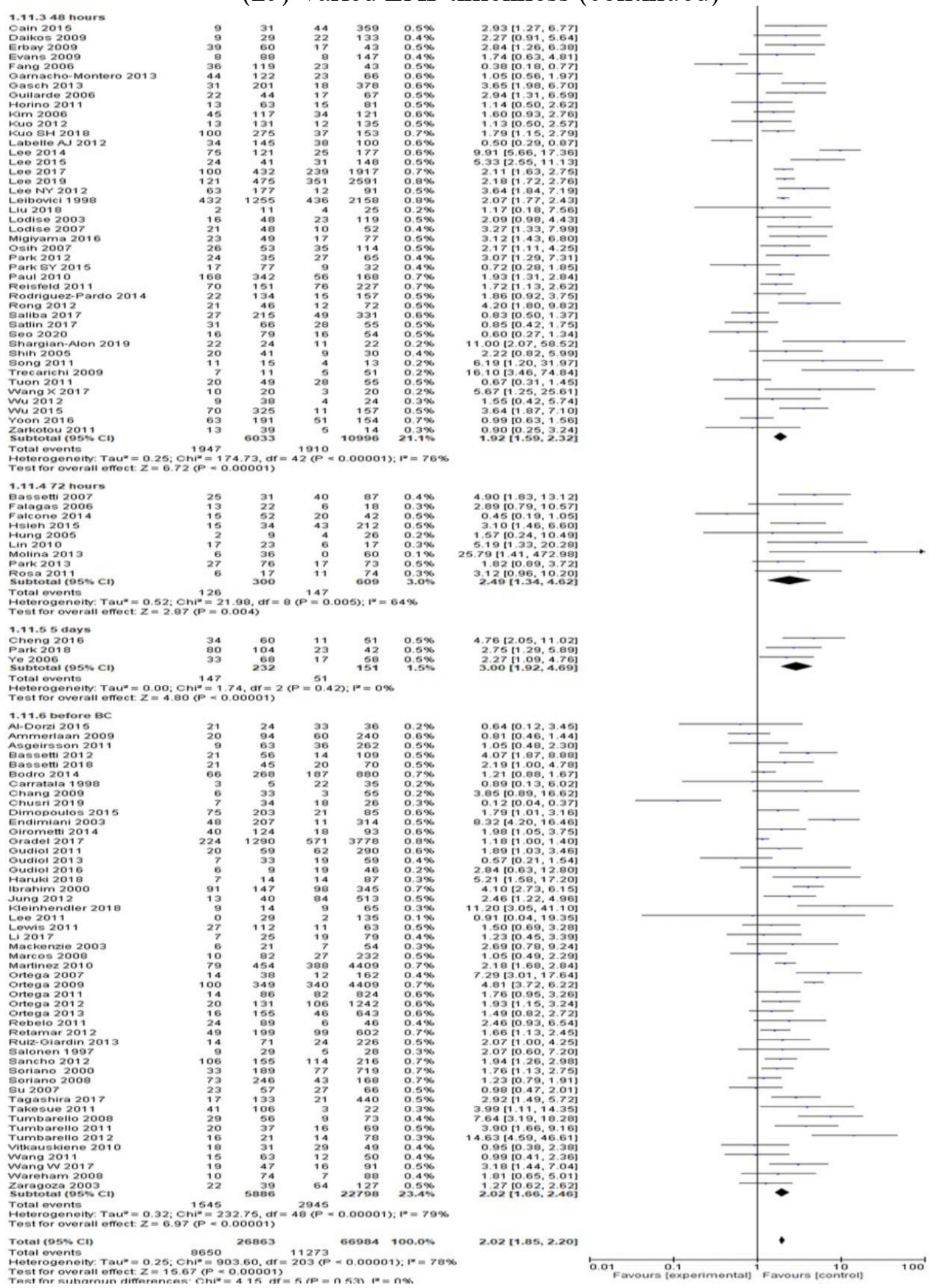
(2I) Antibiotic -resistant microorganism



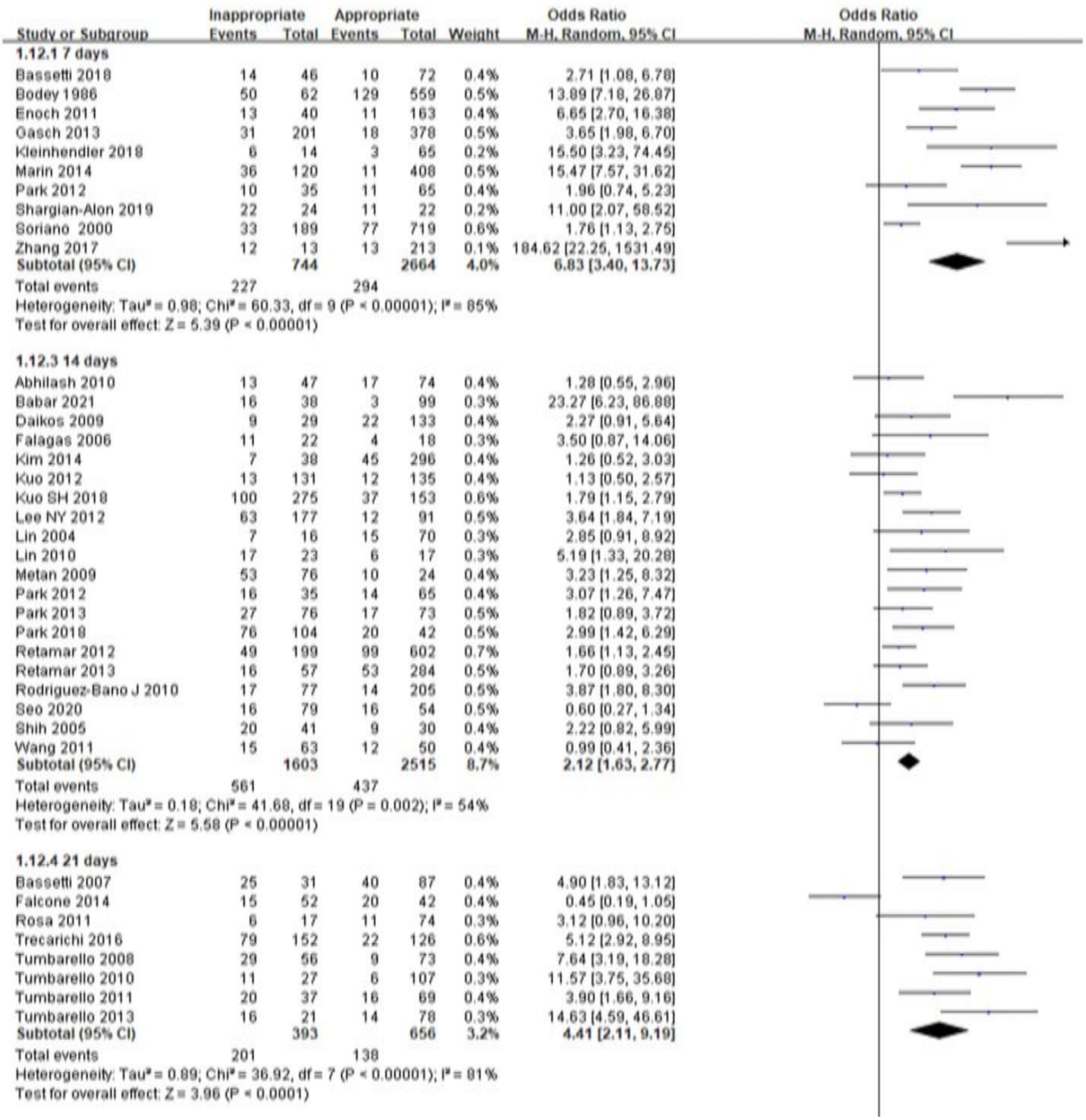
(2J) Varied EAT timeliness



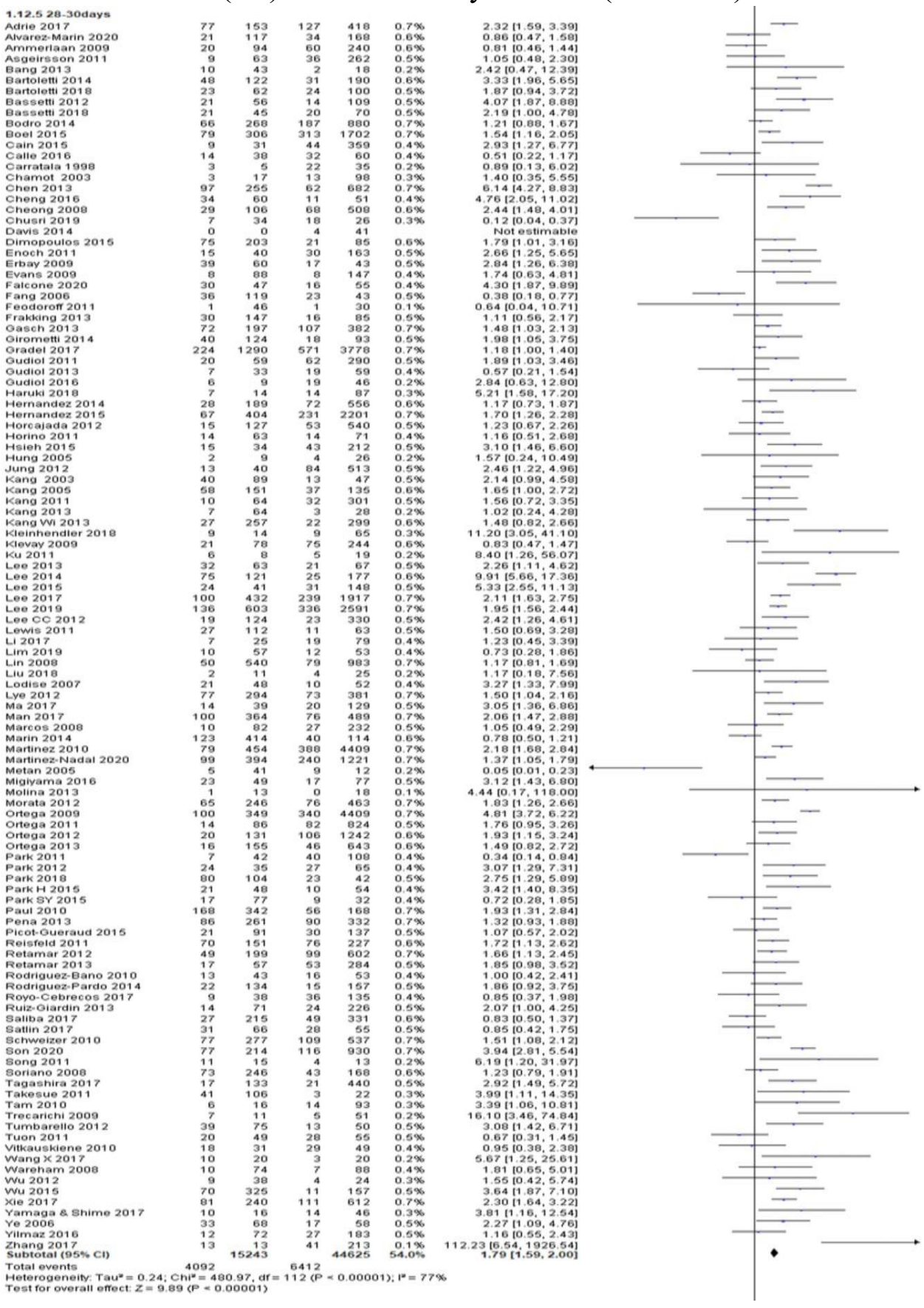
(2J) Varied EAT timeliness (continued)



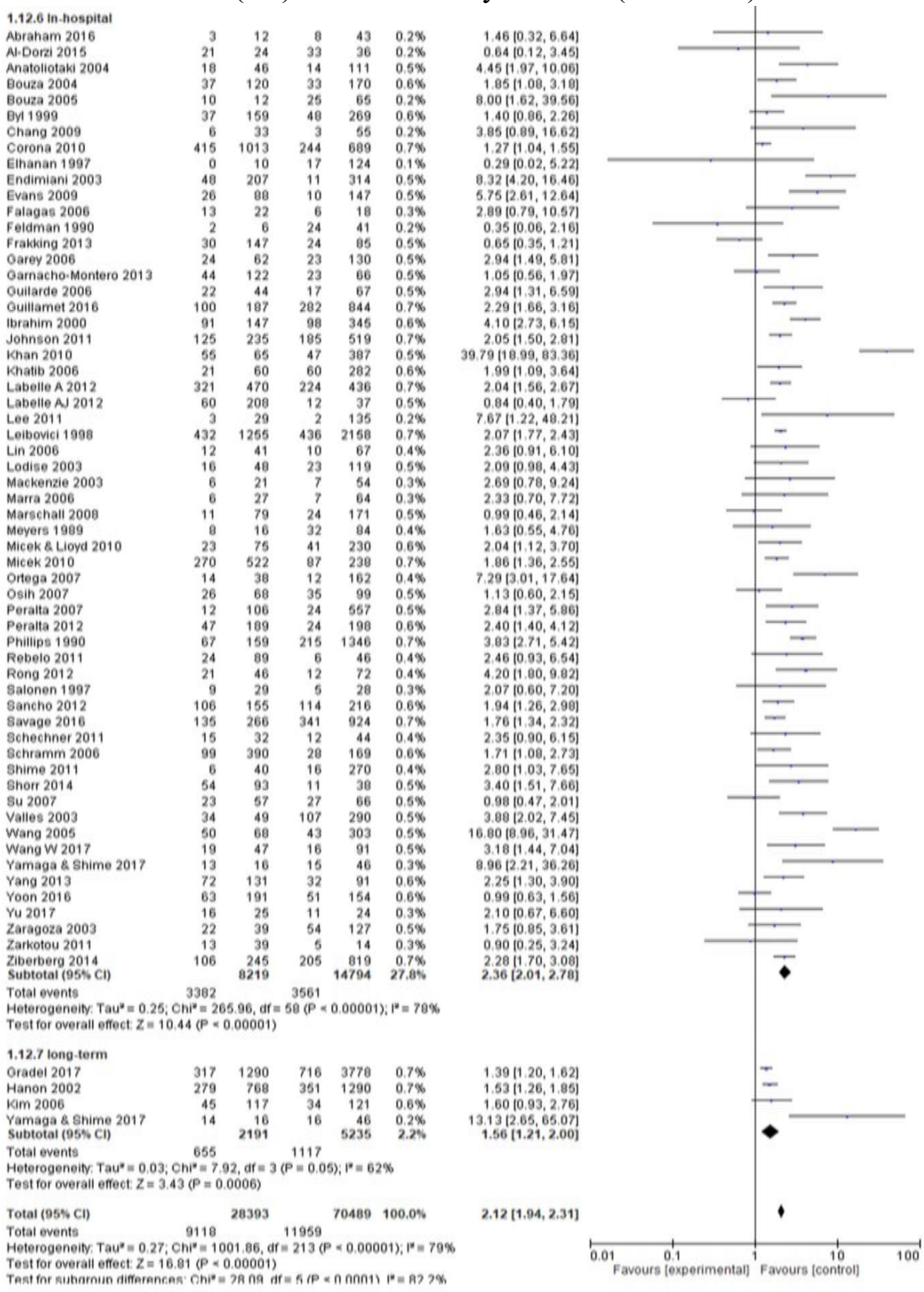
(2K) Varied mortality deadline



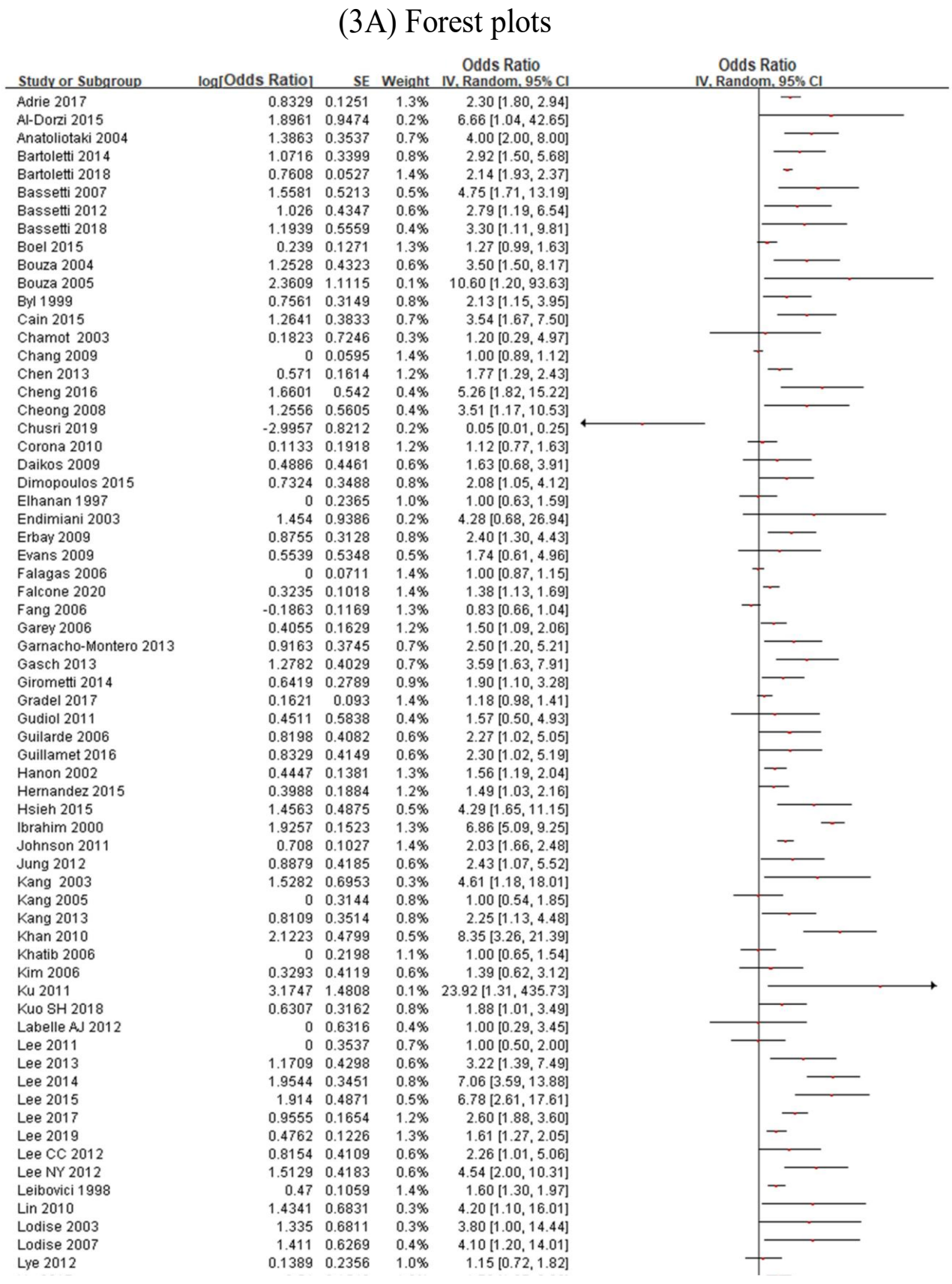
(2K) Varied mortality deadline (continued)



(2K) Varied mortality deadline (continued)



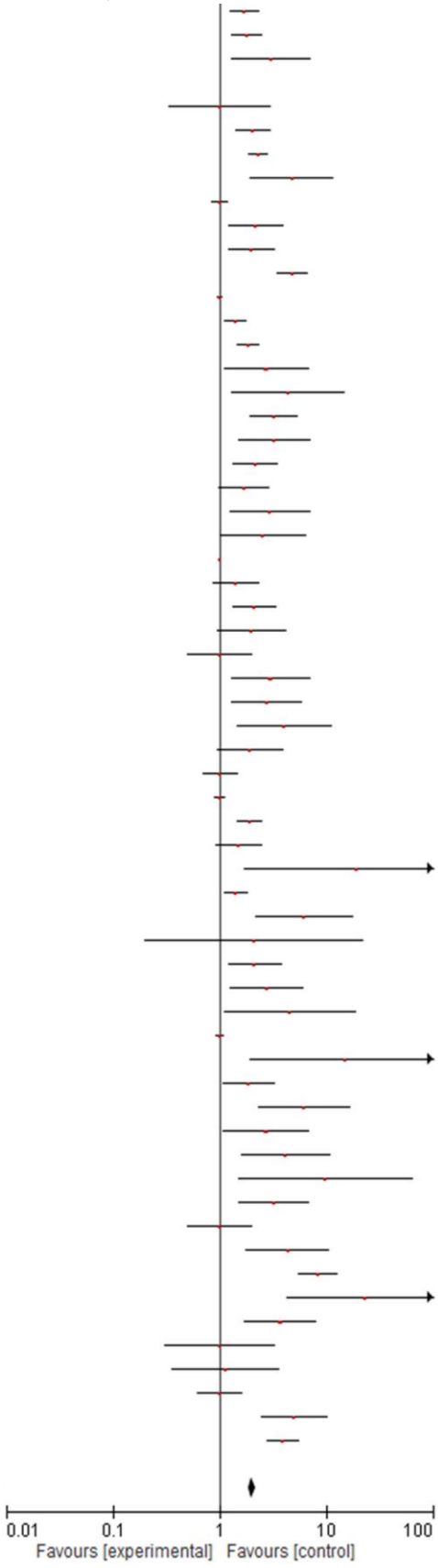
Supplemental Figure 3. Adjusted analyses in overall patients: the forest and funnel plots.



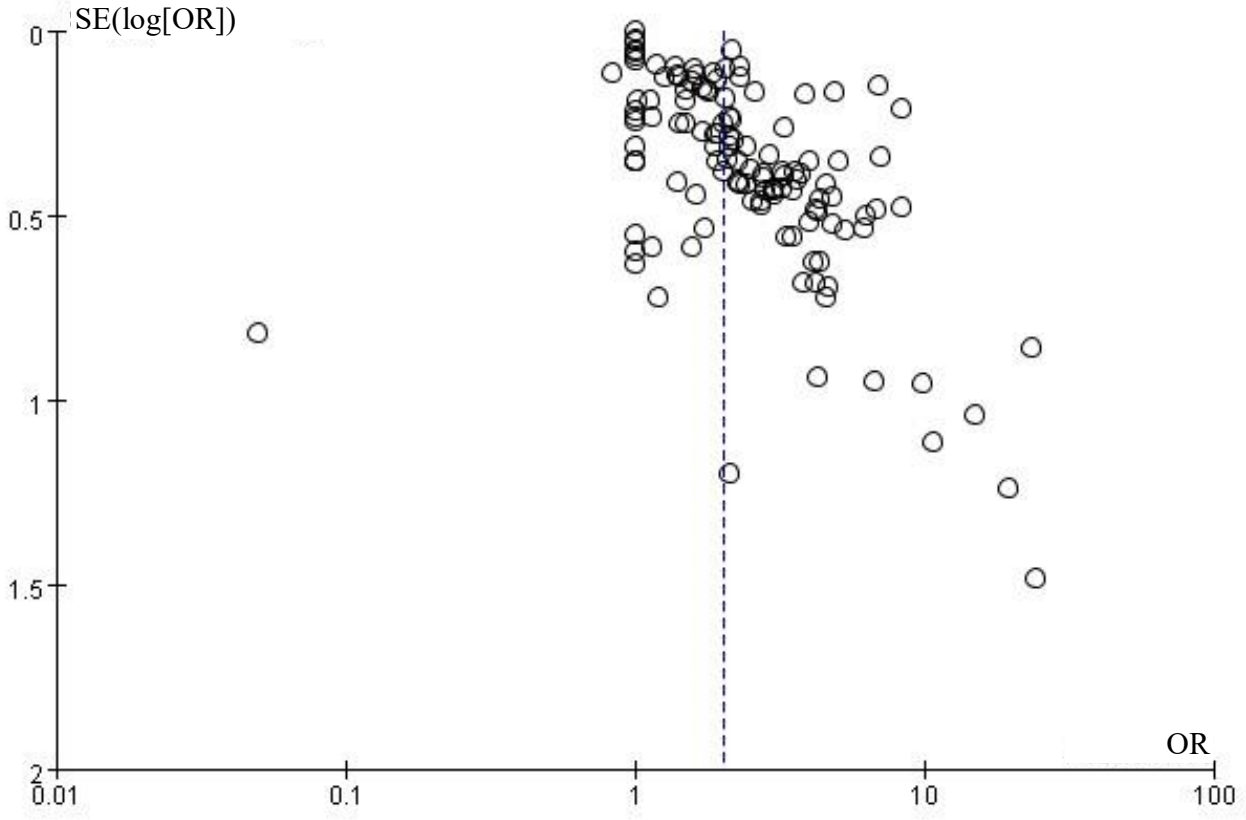
(3A) Forest plots (continued)

Ma 2017	0.54	0.1548	1.2%	1.72 [1.27, 2.32]
Martinez 2010	0.5878	0.166	1.2%	1.80 [1.30, 2.49]
Martinez-Nadal 2020	1.1053	0.434	0.6%	3.02 [1.29, 7.07]
Metan 2005	0	0		Not estimable
Metan 2009	0	0.5504	0.4%	1.00 [0.34, 2.94]
Micek & Lloyd 2010	0.7129	0.1848	1.2%	2.04 [1.42, 2.93]
Micek 2010	0.8329	0.1002	1.4%	2.30 [1.89, 2.80]
Migiyama 2016	1.5602	0.4527	0.6%	4.76 [1.96, 11.56]
Molina 2013	0	0.0829	1.4%	1.00 [0.85, 1.18]
Morata 2012	0.7793	0.2983	0.9%	2.18 [1.21, 3.91]
Ortega 2007	0.6931	0.2522	1.0%	2.00 [1.22, 3.28]
Ortega 2009	1.5748	0.1673	1.2%	4.83 [3.48, 6.70]
Ortega 2011	0	0.0262	1.5%	1.00 [0.95, 1.05]
Ortega 2012	0.3365	0.123	1.3%	1.40 [1.10, 1.78]
Ortega 2013	0.6206	0.1166	1.3%	1.86 [1.48, 2.34]
Park 2012	1.0043	0.4592	0.6%	2.73 [1.11, 6.71]
Park 2013	1.4695	0.6238	0.4%	4.35 [1.28, 14.76]
Park 2018	1.1756	0.2617	1.0%	3.24 [1.94, 5.41]
Park H 2015	1.1756	0.3929	0.7%	3.24 [1.50, 7.00]
Paul 2010	0.7655	0.2412	1.0%	2.15 [1.34, 3.45]
Pena 2013	0.5306	0.2759	0.9%	1.70 [0.99, 2.92]
Peralta 2007	1.0919	0.4433	0.6%	2.98 [1.25, 7.10]
Peralta 2012	0.94	0.4645	0.6%	2.56 [1.03, 6.36]
Rebelo 2011	0	0.0051	1.5%	1.00 [0.99, 1.01]
Reisfeld 2011	0.3436	0.2523	1.0%	1.41 [0.86, 2.31]
Retamar 2012	0.7514	0.2341	1.0%	2.12 [1.34, 3.35]
Retamar 2013	0.6931	0.3798	0.7%	2.00 [0.95, 4.21]
Rodriguez-Bano 2010	0	0.3537	0.7%	1.00 [0.50, 2.00]
Rodriguez-Bano J 2010	1.0986	0.4267	0.6%	3.00 [1.30, 6.92]
Rodriguez-Pardo 2014	1.0188	0.3868	0.7%	2.77 [1.30, 5.91]
Rong 2012	1.3863	0.5181	0.5%	4.00 [1.45, 11.04]
Sancho 2012	0.6575	0.3563	0.7%	1.93 [0.96, 3.88]
Savage 2016	0.0198	0.1921	1.1%	1.02 [0.70, 1.49]
Schechner 2011	0	0.0538	1.4%	1.00 [0.90, 1.11]
Schramm 2006	0.6523	0.1328	1.3%	1.92 [1.48, 2.49]
Schweizer 2010	0.4055	0.2494	1.0%	1.50 [0.92, 2.45]
Shargian-Alon 2019	2.9565	1.2407	0.1%	19.23 [1.69, 218.82]
Shorr 2014	0.3492	0.13	1.3%	1.42 [1.10, 1.83]
Son 2020	1.8148	0.533	0.5%	6.14 [2.16, 17.45]
Song 2011	0.7419	1.1997	0.1%	2.10 [0.20, 22.05]
Soriano 2000	0.7561	0.2885	0.9%	2.13 [1.21, 3.75]
Tagashira 2017	1.0225	0.3997	0.7%	2.78 [1.27, 6.09]
Takesue 2011	1.514	0.7248	0.3%	4.54 [1.10, 18.81]
Tam 2010	0	0.0316	1.5%	1.00 [0.94, 1.06]
Trecarichi 2009	2.7054	1.0396	0.2%	14.96 [1.95, 114.78]
Trecarichi 2016	0.6259	0.2801	0.9%	1.87 [1.08, 3.24]
Tumbarello 2008	1.8278	0.501	0.5%	6.22 [2.33, 16.61]
Tumbarello 2011	1.0043	0.4731	0.5%	2.73 [1.08, 6.90]
Tumbarello 2012	1.4279	0.4856	0.5%	4.17 [1.61, 10.80]
Tumbarello 2013	2.2875	0.9535	0.2%	9.85 [1.52, 63.84]
Valles 2003	1.1725	0.3846	0.7%	3.23 [1.52, 6.86]
Wang X 2017	0	0.3537	0.7%	1.00 [0.50, 2.00]
Wu 2012	1.4697	0.4542	0.6%	4.35 [1.79, 10.59]
Xie 2017	2.1136	0.2123	1.1%	8.28 [5.46, 12.55]
Yamaga & Shime 2017	3.1465	0.8556	0.2%	23.25 [4.35, 124.39]
Yang 2013	1.3124	0.3882	0.7%	3.72 [1.74, 7.95]
Ye 2006	0	0.5976	0.4%	1.00 [0.31, 3.23]
Yu 2017	0.1275	0.5849	0.4%	1.14 [0.36, 3.57]
Zaragoza 2003	0	0.2439	1.0%	1.00 [0.62, 1.61]
Zhang 2017	1.6094	0.3537	0.7%	5.00 [2.50, 10.00]
Ziberberg 2014	1.3538	0.1709	1.2%	3.87 [2.77, 5.41]
Total (95% CI)			**100.0%**	**2.02 [1.86, 2.20]**

Heterogeneity: Tau² = 0.13; Chi² = 1523.69, df = 124 (P < 0.00001); I² = 92%
Test for overall effect: Z = 16.17 (P < 0.00001)

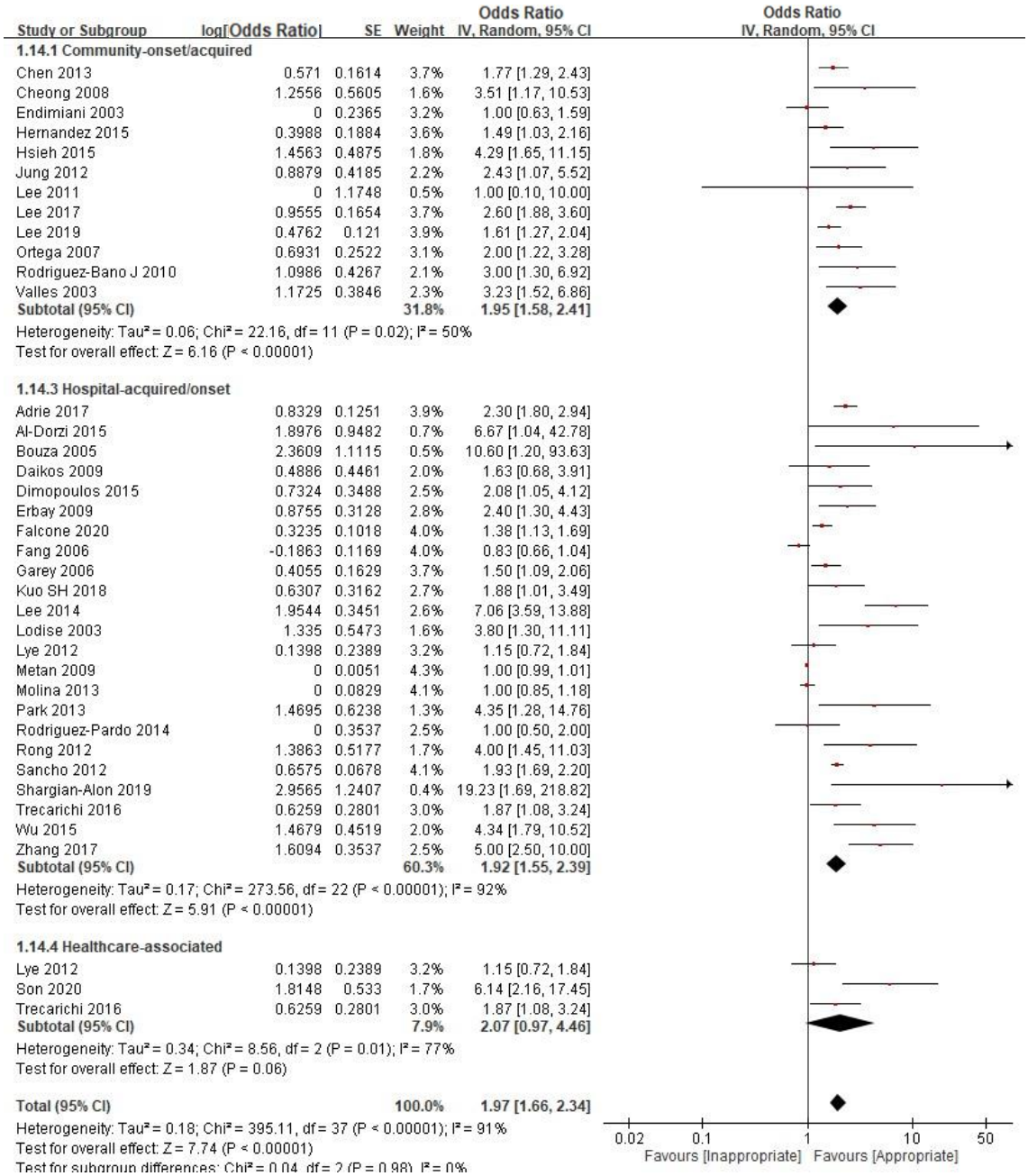


(3B) Funnel plots

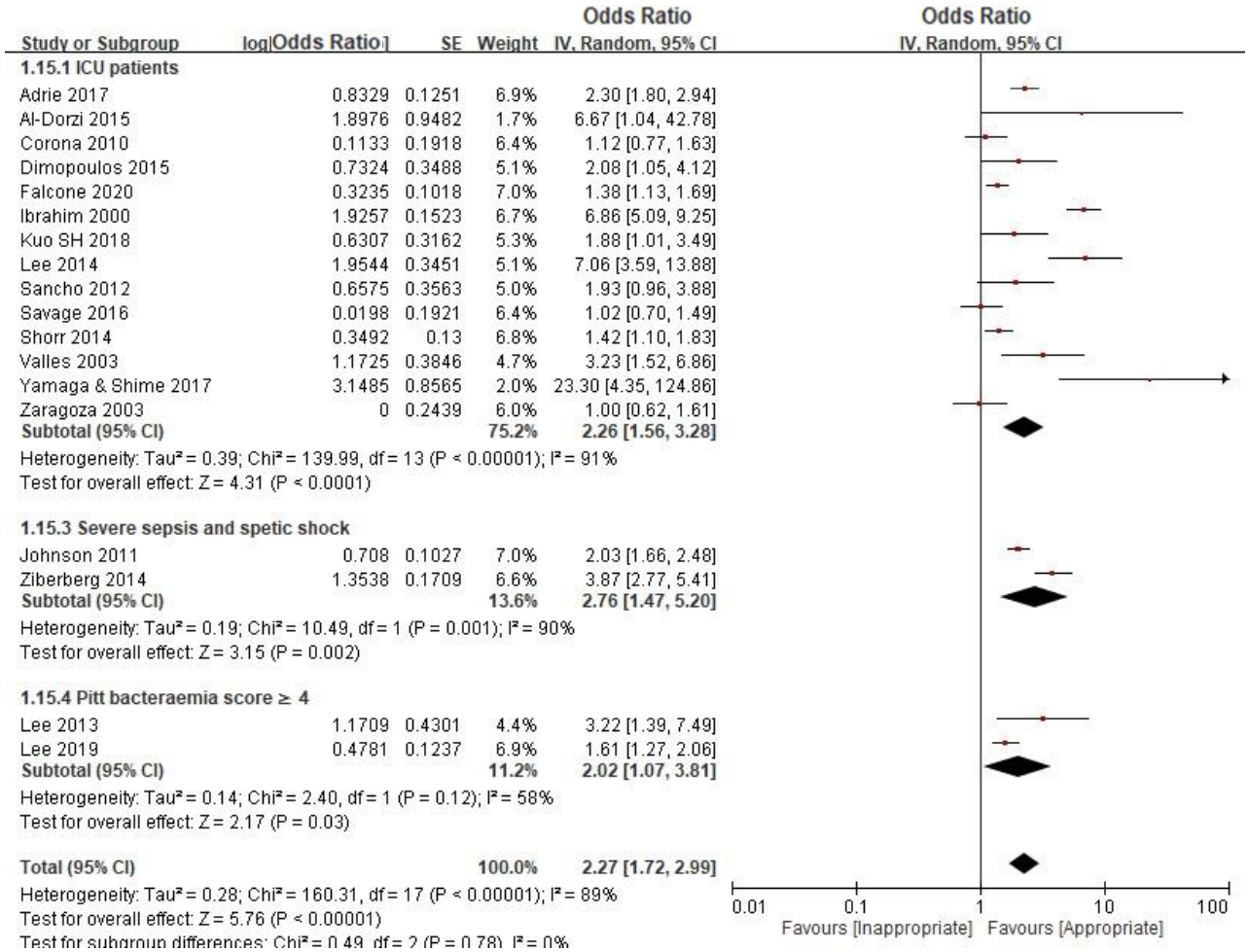


Supplemental Figure 4. Adjusted analyses in subgroup patients: (4A) acquisition places; (4B) bacteraemia severity; (4C) specific comorbidities or conditions; (4D) bacteraemia sources; (4E) Gram-positive cocci; (4F) Enterobacteriaceae; (4G) glucose non-fermentative rods; (4H) antibiotic-resistant microorganisms; (4I) varied timeliness of empirical antimicrobial therapy (EAT); and (4J) varied mortality timeliness

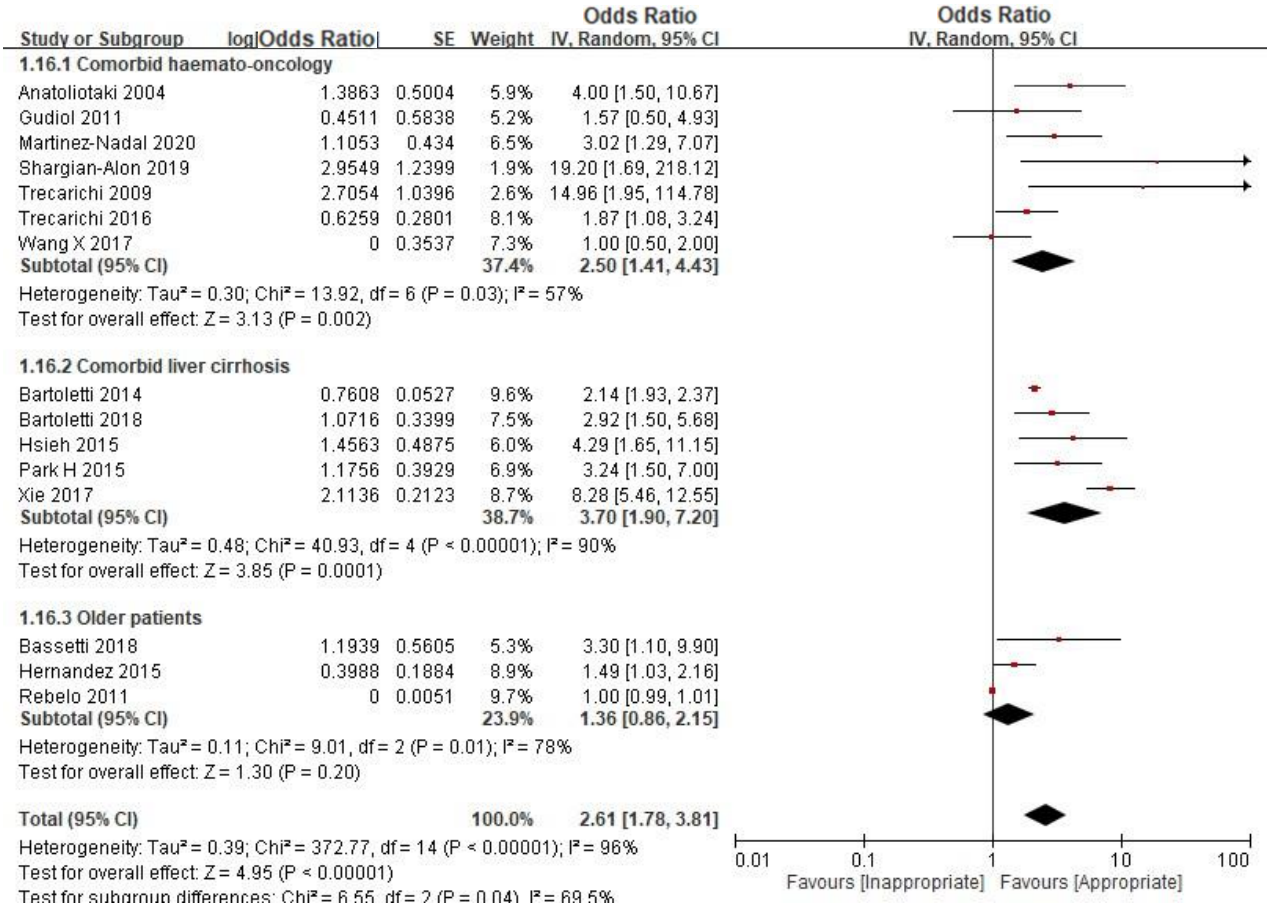
(4A) Acquisition places



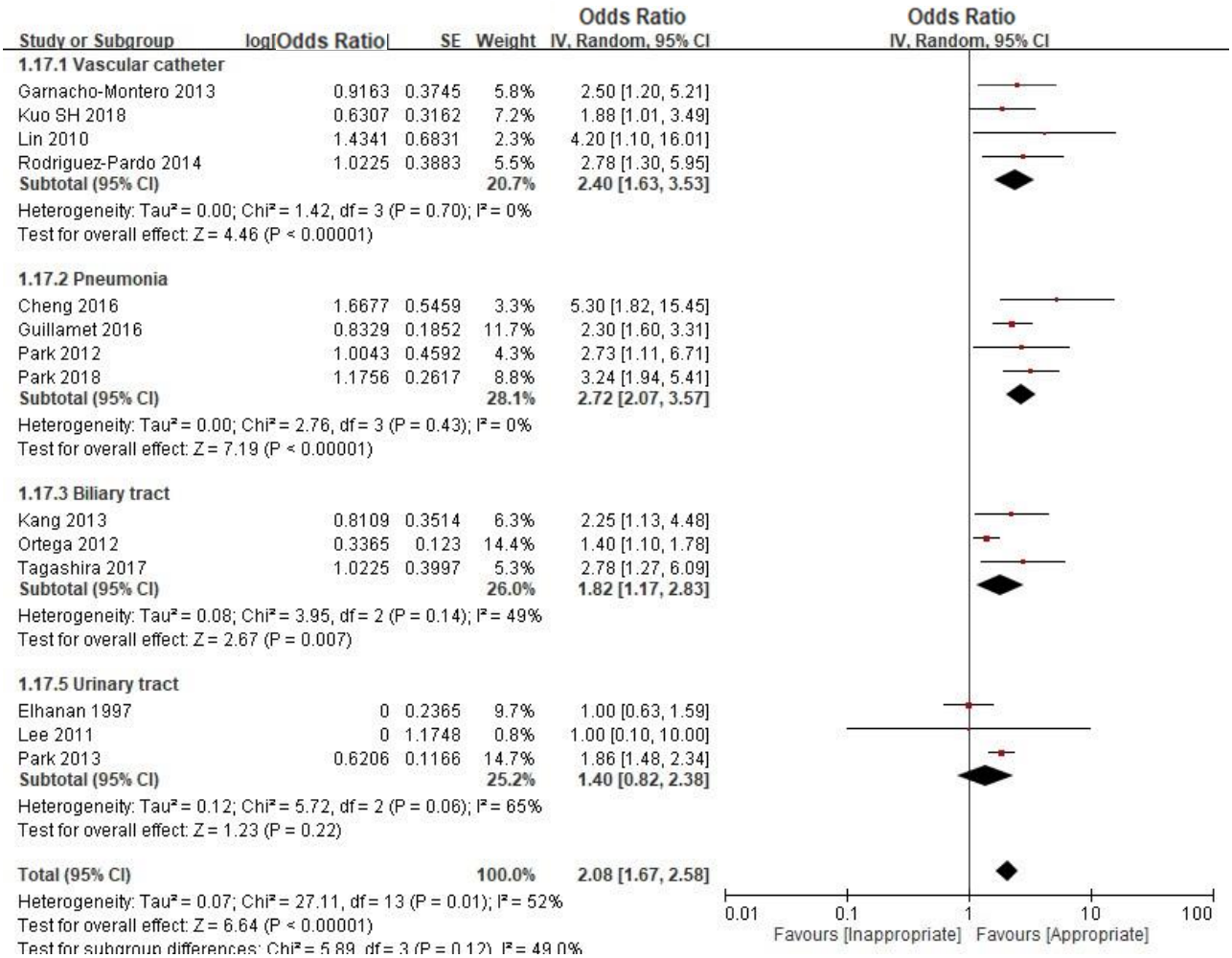
(4B) Bacteraemia severity



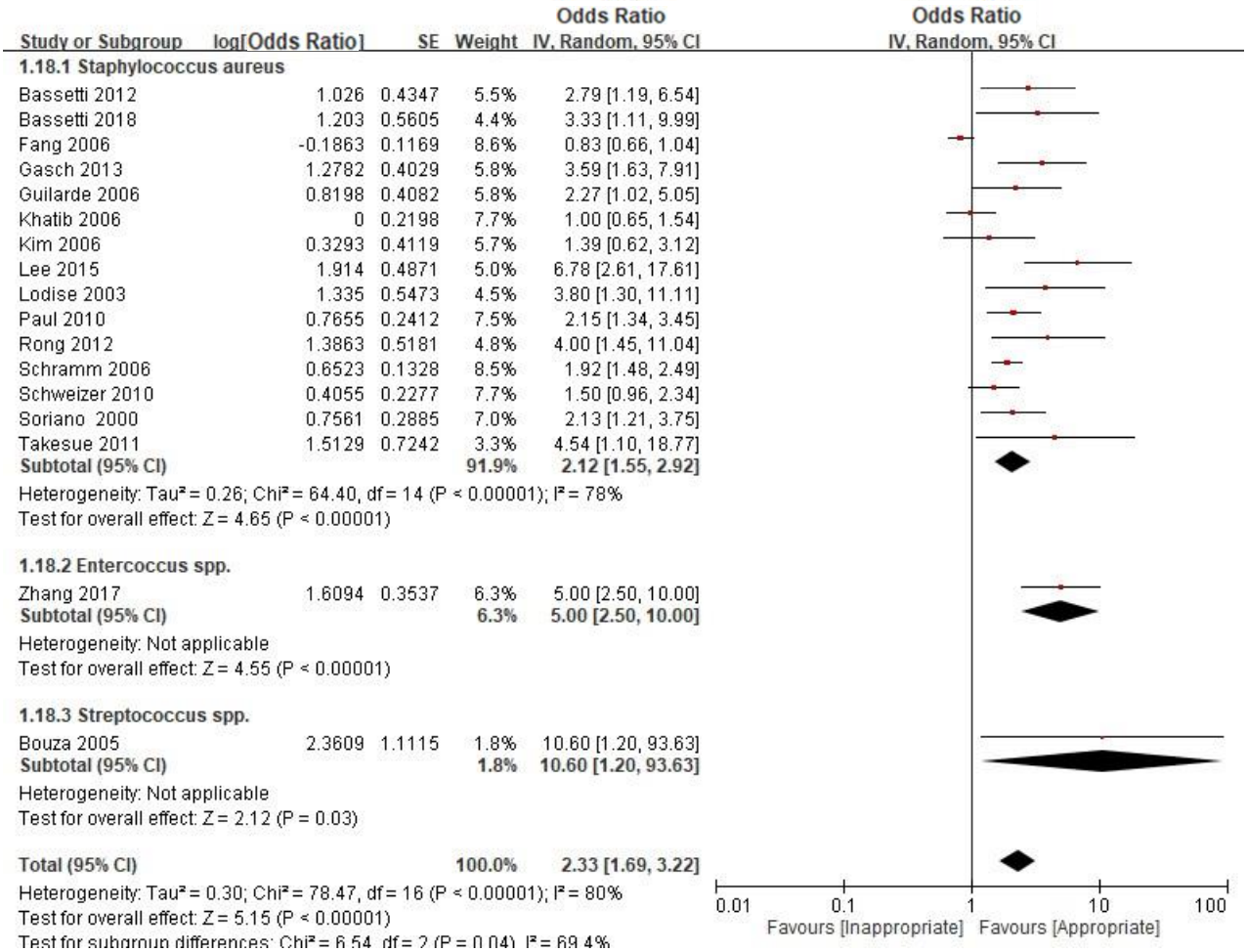
(4C) Specific comorbidities or condition



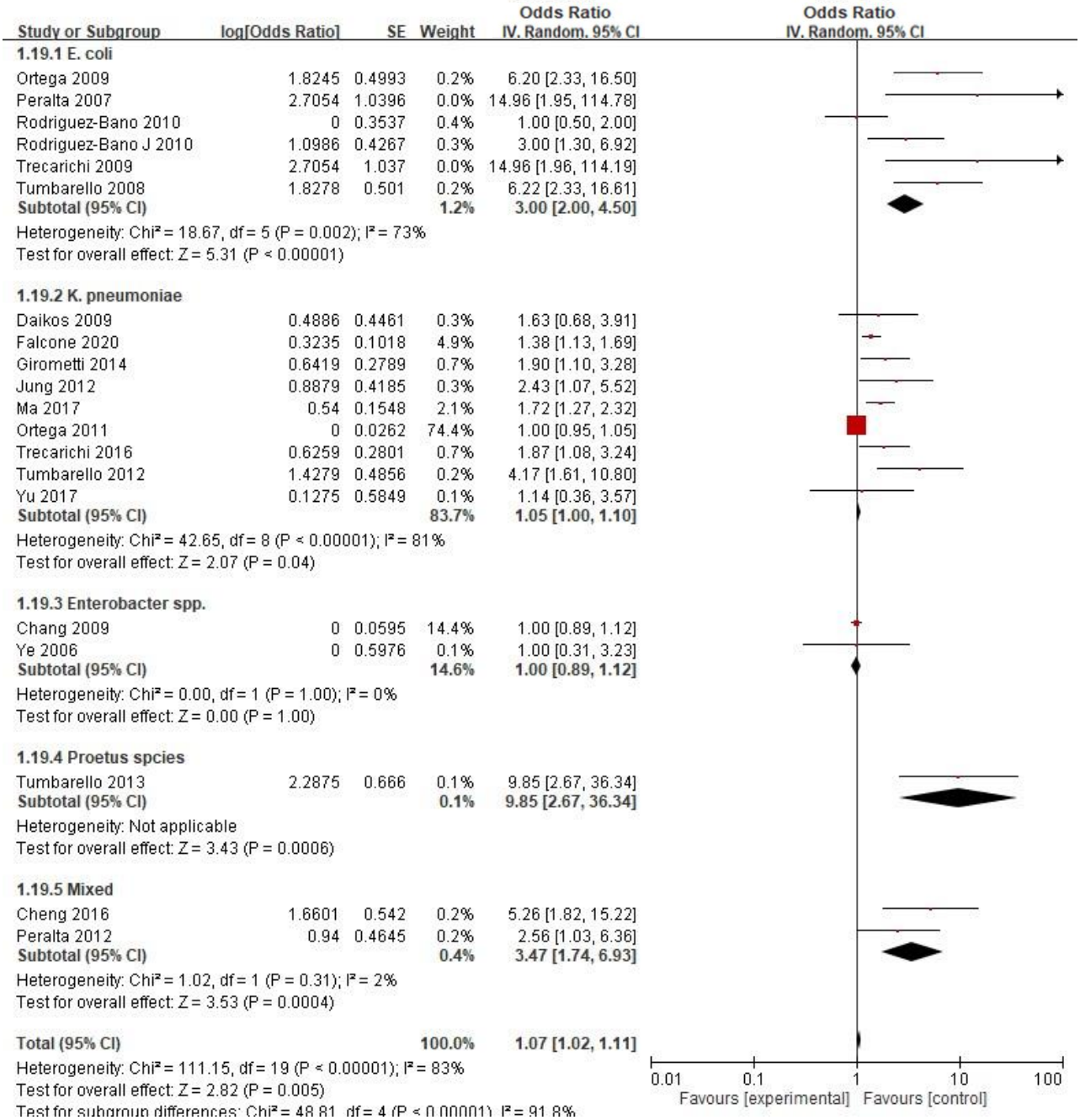
(4D) Bacteraemia source



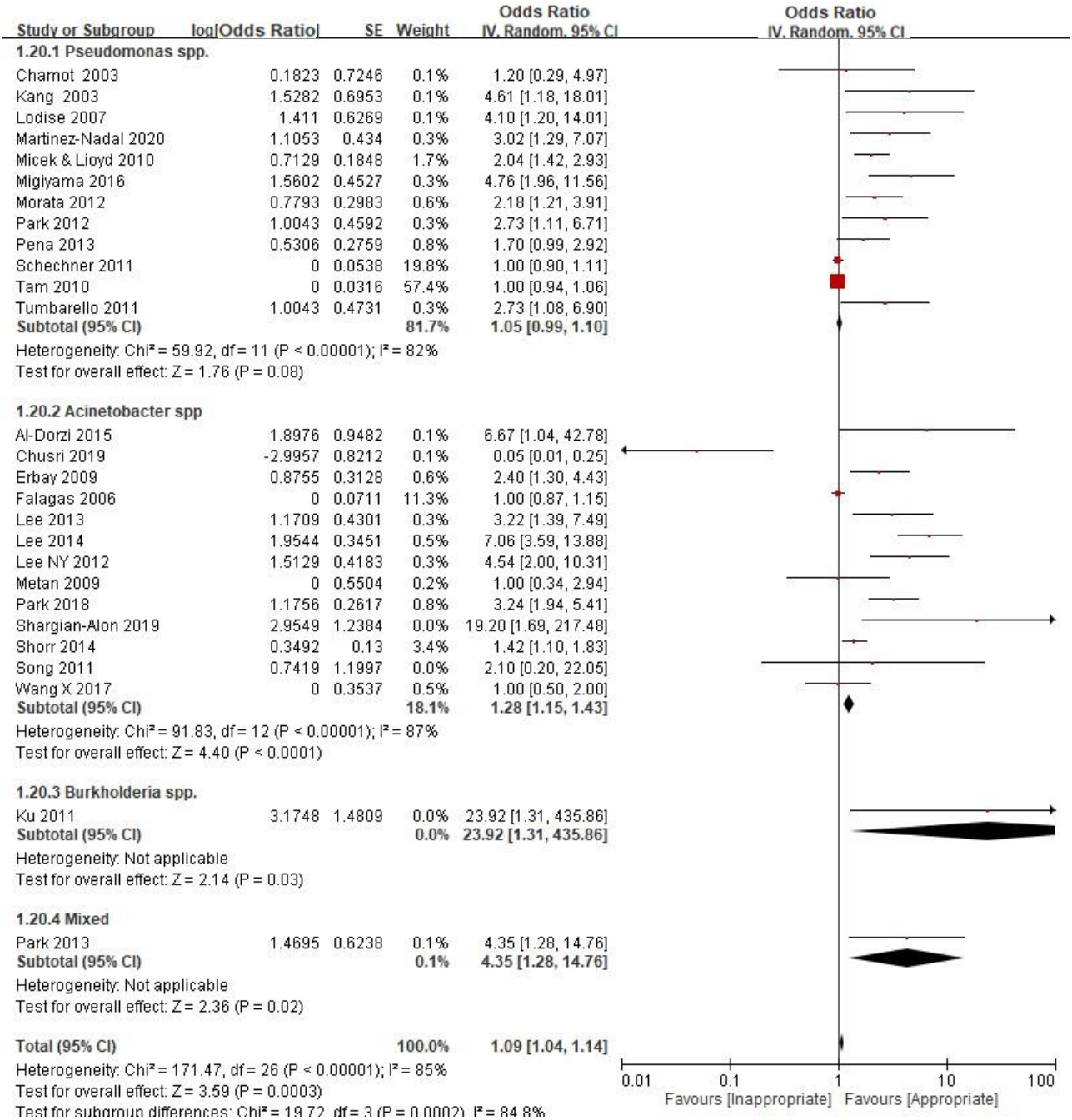
(4E) Gram-positive cocci



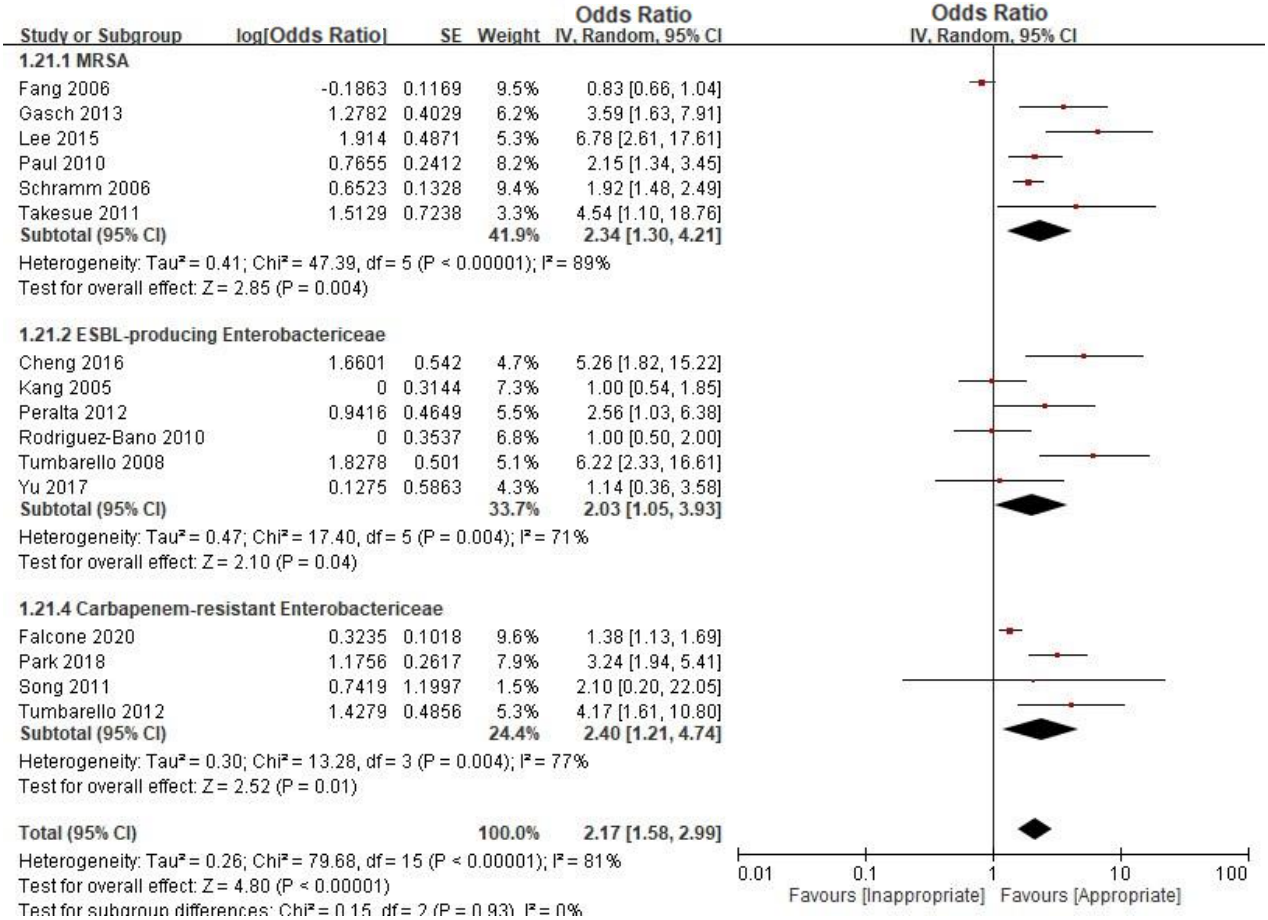
(4F) Enterobacteriaceae



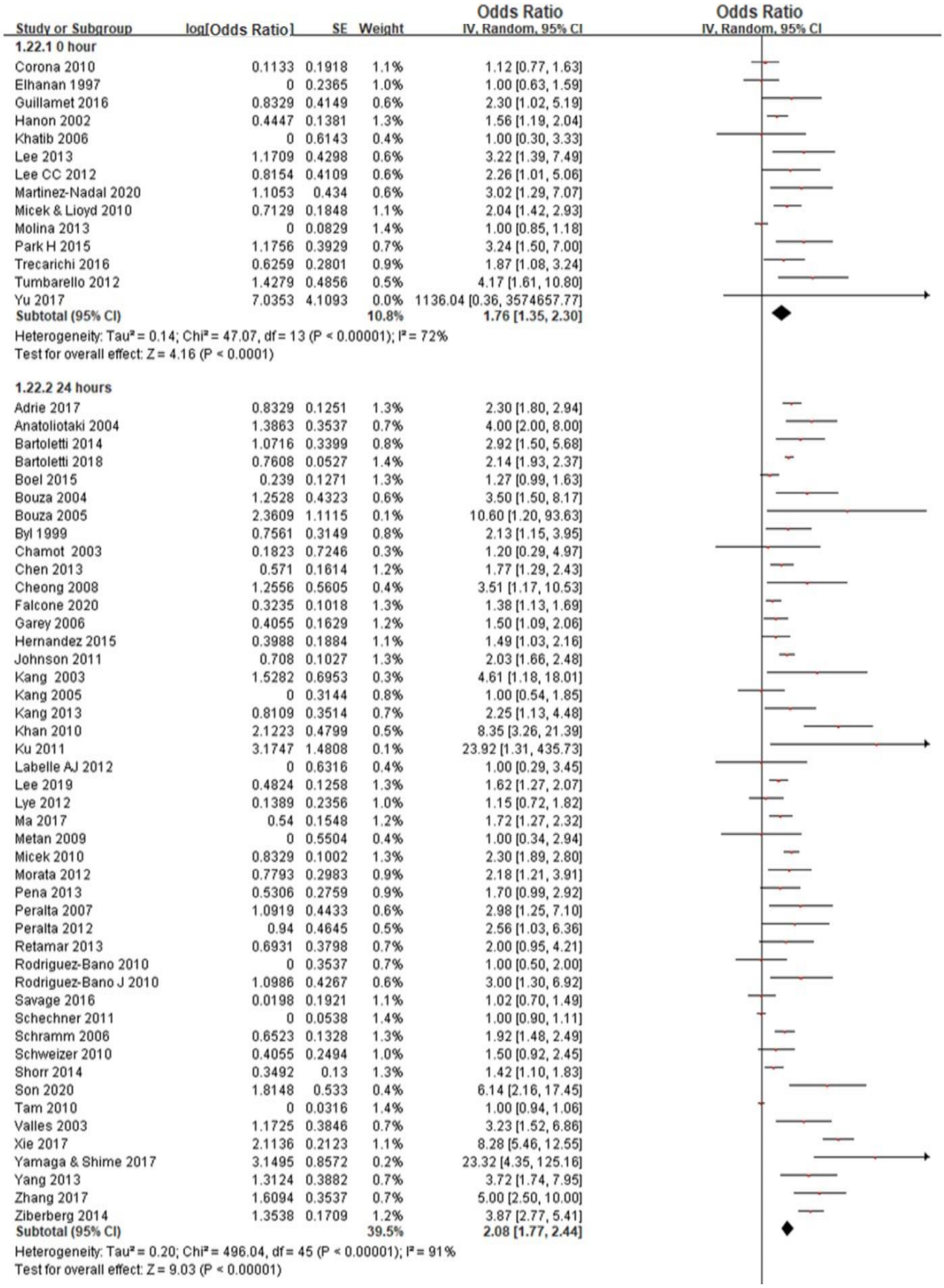
(4G) Glucose non-fermentative rods



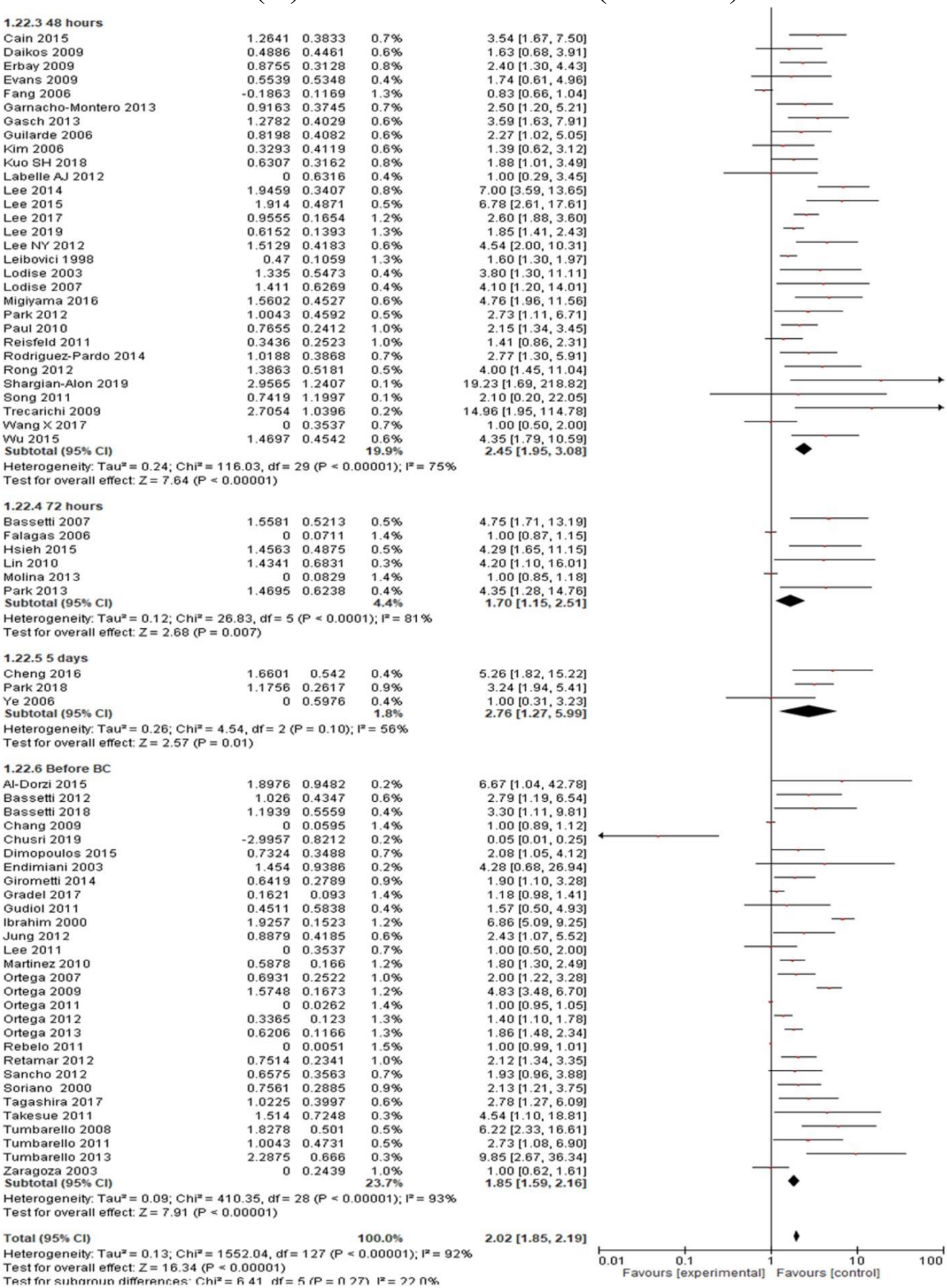
(4H) Antibiotic-resistant microorganism



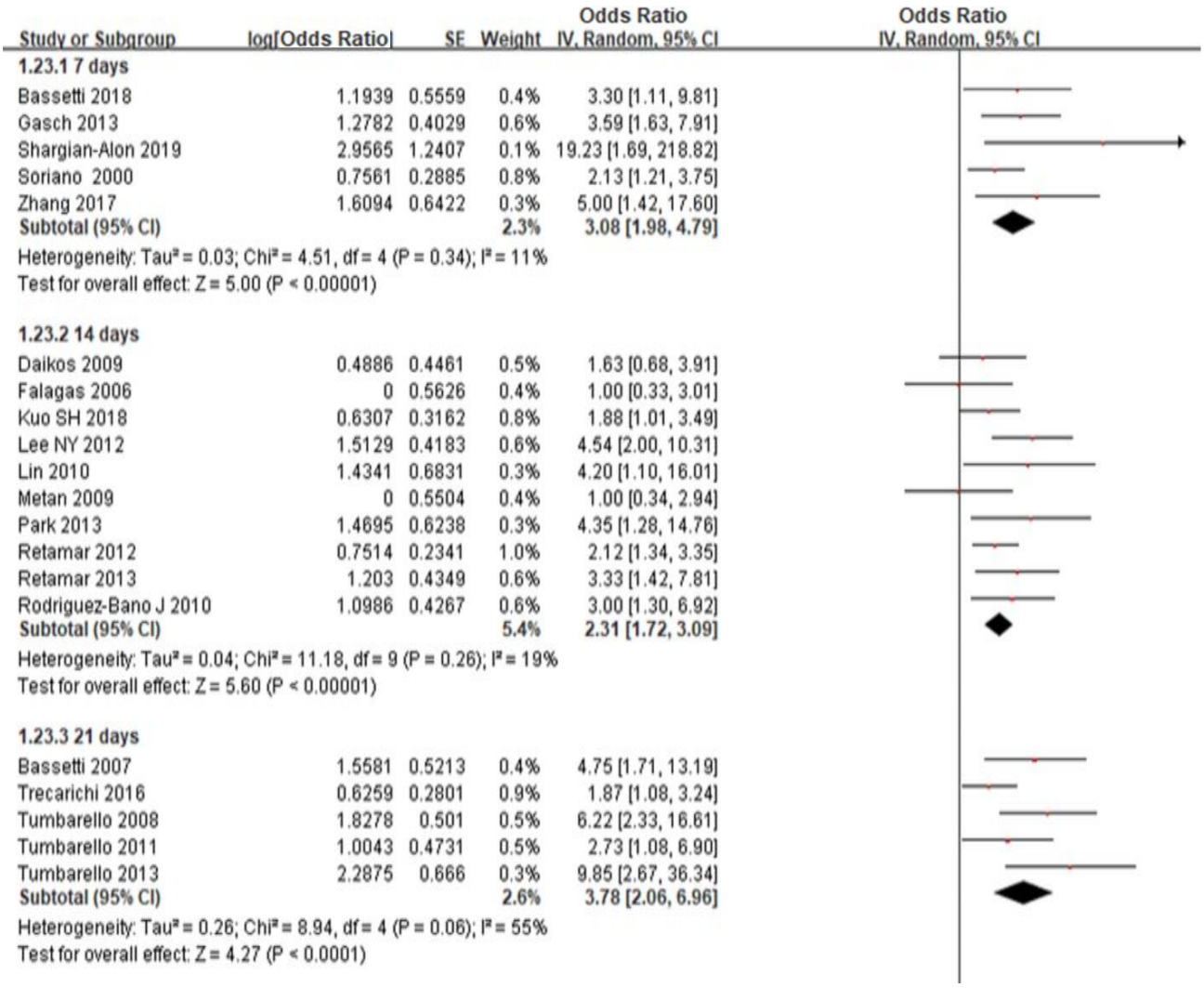
(4I) Varied EAT timeliness



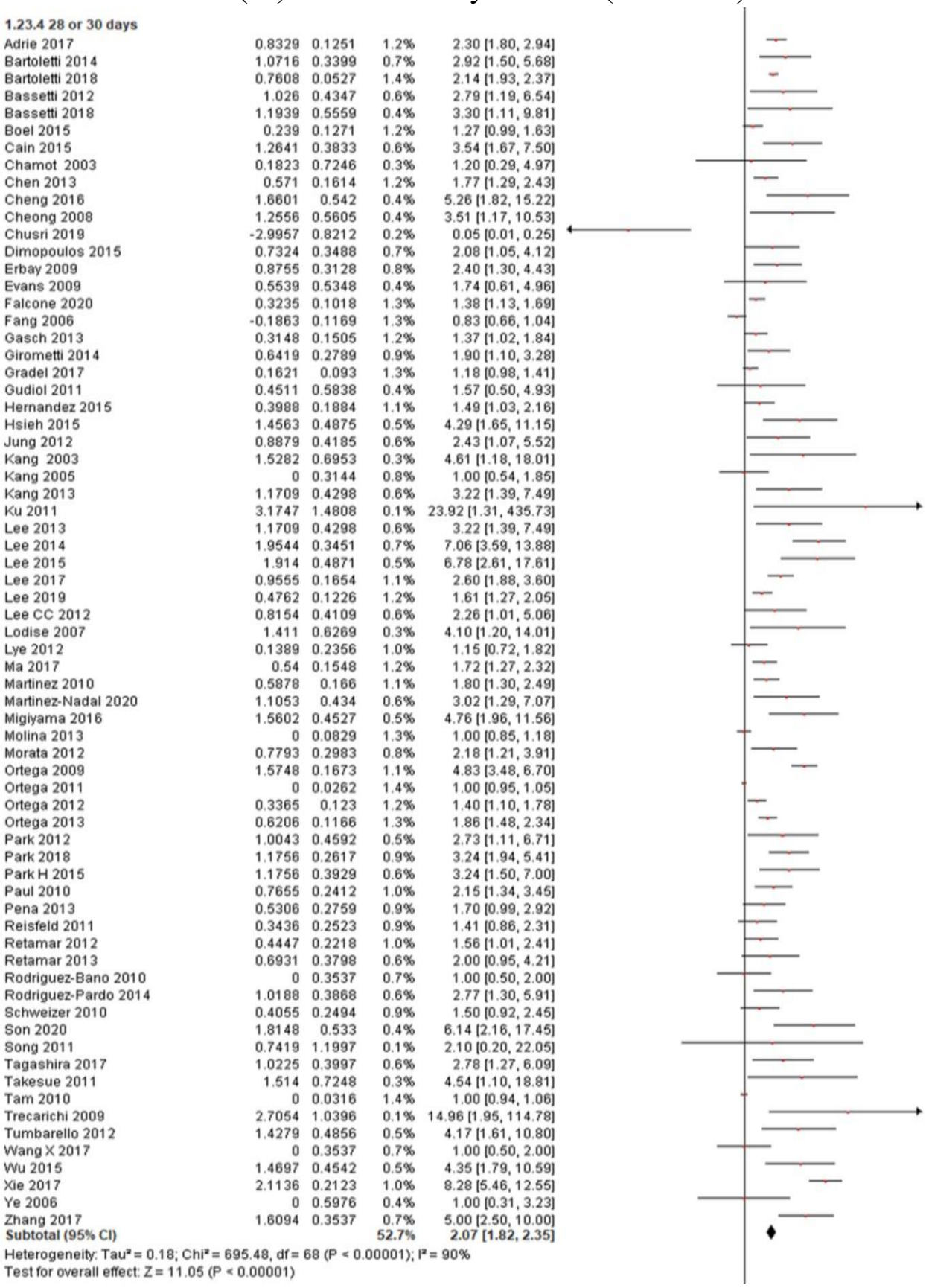
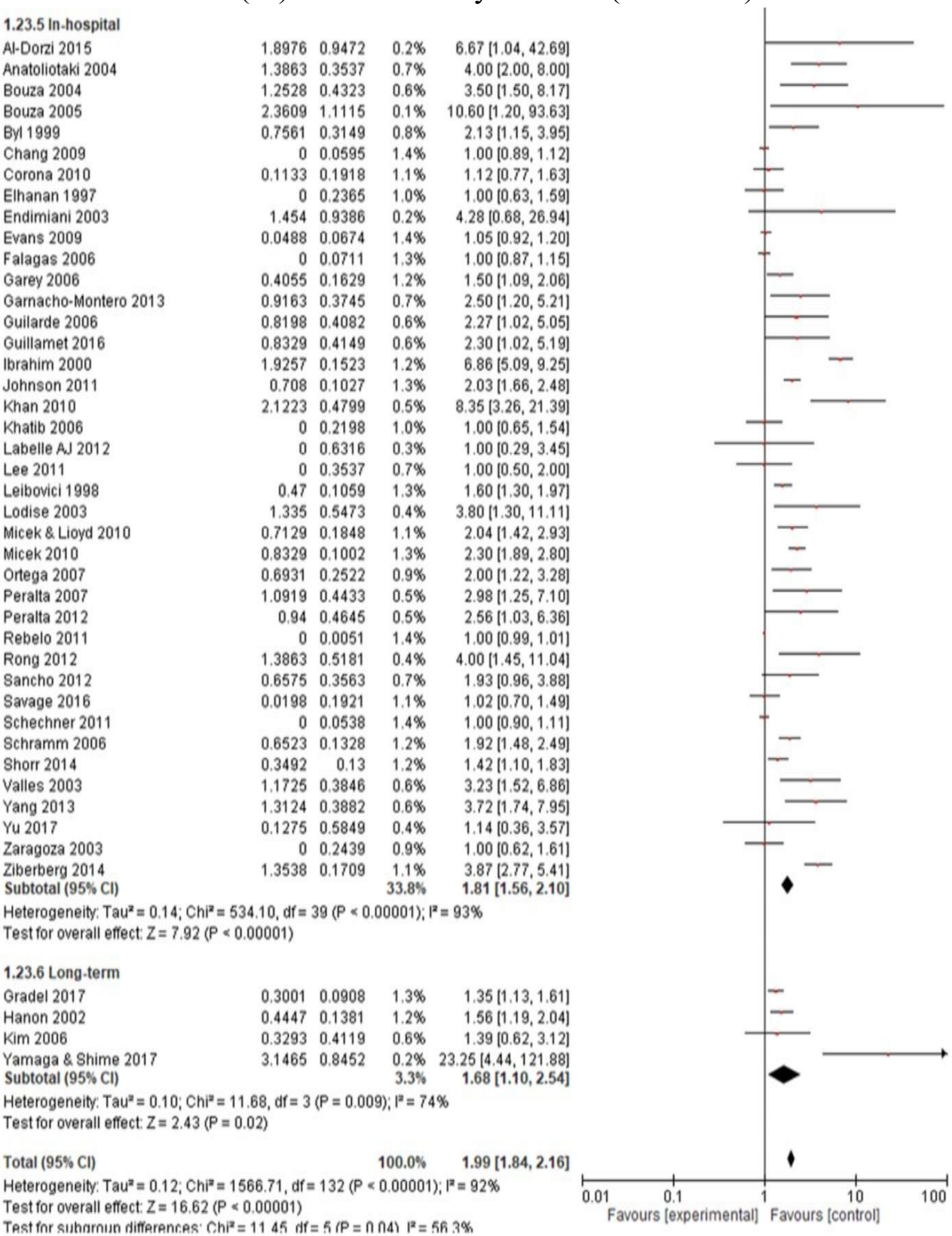
(4I) Varied EAT timeliness (continued)



(4J) Varied morality deadline



(4J) Varied morality deadline (continued)



(4J) Varied morality deadline (continued)